Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 — Unit 1 · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_01 · Unit 1

1 UE26MA141A · Unit_01: Unit 1

This unit master organizes source material for Engineering Mathematics - I, Unit_01, officially titled Unit 1. It includes 17 video transcripts and 11 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26MA141A · Unit_01: Unit 1

Official catalogue entry: UE26MA141A, Unit_01 (Unit 1). Bucket folder present: yes. Videos: 17; slide decks: 11.

EV0000001

2 Continuity and differentiability of a function of two independent variable

Study focus: Continuity and differentiability of a function of two independent variable. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we'll be discussing continuity, history of partial derivative, definition of partial derivative, higher order partial derivative, and its geometric interpretation.

Key details

- In the previous session,.
- We defined what is the continuity of a function of single variable.

- A similar thing can be done for a function of two independent variables.
- We'll solve some examples so that the definition will be a little more clear to you.
- A function is continuous at a point.
- So geometrically it means that there is no hole or break in the graph.
- Unlike see for a function of single variable, what we said is a function is said to be discontinuous if there is a gap.
- But for a function of bivariate, I need to be in three-dimensional plane.
- I have x, y, and z axis.
- My surface would look something like this.
- But there is a hole.
- I'll show you how it looks.
- I can't call that as a break.
- A break can be considered only for a curve.
- But you need to be careful about something.
- A function is not continuous doesn't imply that the limit does exist.
- is not a condition, right?
- So that means even if the limit exists, the function need not be continuous.
- I know things are not that clear.
- If you observe carefully, as I go closer to 0, see if I come closer to 0, there is a folding that is happening.
- So that means there is actually a gap that is there at 0, 0.
- I need to check if this function is continuous or the curve must be very smooth.
- That's what I mean by continuous, right?
- How do I check if the function is continuous?
- I need to check the limit as xy.
- x y approaching to zero of this function 3x square y divided by x square plus y square this must be equal to f of 0 comma 0.
- That's the condition.
- But now is it satisfied is the question.
- For this function, what is the value of f of 0,0?
- If I substitute, I will get 3 into 0 into 0 divided by 0 plus 0, which is obviously in the form of 0 by 0, which is an indeterminate form.
- And no matter whatever the value I get in the left hand side, I cannot compare this with an indeterminate form.
- So this function is not continuous, whatever the function I have here is not continuous because the function is not defined at the origin.
- Limit might exist.
- But no matter whatever the value of this limit is, I cannot equate this to an indeterminate value.
- So function is not continuous because the value of the function is not defined at the origin.
- Is it clear?
- It's the same thing that we did yesterday.

Example or application from the lecture

- So let's take an example.
- Before we attempt to find the left hand side, let's check the value of f of 00.
- So I hope you understood the first example.

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EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

Let's see, what is this example now?

Key details

- You look at the graph carefully.
- This is the same function we took yesterday to explain the limits in the previous session, right?
- Limit for this.
- As, the limit does not exist, right?
- of this function, which is 2xy divided by 3x square plus y square.
- And we calculated along different paths we took y equal to 0 and then we took y is equal to x.
- x equal to 0.
- You take any path, the value that we get is different.
- Also the function, what is the value of the function at 0 comma 0?
- It will be 0 by 0, which is again an indeterminate form.
- Look at the graph a little bit carefully.
- At zero, there is actually a crunch or whatever you want to call it.
- It's like a boundary and then it suddenly drops down at 0, 0, and then of course at every other point the function exists.
- And as usual, as I mentioned in the previous lecture, it is very easy for us to check if the function is not continuous.
- If you want to check if the function is continuous, the best way to do that is look at the graph.
- Next, I want to go a little bit or talk a little bit about the history of partial differentiation.
- Who invented all these things?
- At least if not interested.
- Sometimes you want to curse people, right?
- You want to say why did he invent?
- Why did Newton invent so that I have to learn?
- Why did Jacobi do this?
- Why did Leibniz do this?
- but at least we should be grateful to few people who made our life much easier by inventing something that is very, very useful.
- It's only that you're hearing or it's been named differently, right?
- I hope you remember product rule.
- And all of you were very comfortable with product rule because things were very simple.

- Say I have so I have say x q.
- The product rule says that you keep the first function as it is, and you differentiate the second.
- Plus, you keep the second function as it is, and then you differentiate the first.
- And all of you were very comfortable when they define product rule.
- But did you question yourself that when I'm differentiating this, why am I not doing anything to this function here?
- It is just k cos x.
- You will not worry about whatever that constant is.
- The next time what you did, so let me just erase it.
- So the next time what you did is so what I did derivative of x cube into sine x.
- So in the first case I took this as K.
- And I treated that as k.
- So for the next term, that is the second term, what you did?
- You took sine x to be some constant.

Example or application from the lecture

- I took an example of a bivariate function.
- What does this product rule say is suppose I want to differentiate function, which is actually product of two functions.
- Let's take this.

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EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

And then so you treated that as K1 and you differentiated XQ which is 3x square.

Key details

- Is it clear whatever I'm talking?
- So partial differentiation also means the same thing.
- It's not something new.
- D of uv is u d v plus v d u, right?
- So Leibniz was the first person to introduce partial differentiation.
- In fact he in his letter to Lobithal he used the symbol delta.
- for partial derivative with respect to x.
- But later he didn't continue his work on partial derivatives.
- Legendary what he did is he actually defined the modern partial derivative.
- The partial derivative of a function.
- Then the partial derivative of this function is denoted by double.
- The symbol is not delta, this is dou.
- That's a notation.
- You have z, which is a function of more than one variable, and I'm going to differentiate that with respect to x by treating all other variables that are there with me as constant.

- That's what I'm going to do.
- Is the definition clear?
- So the re the very new concept of partial differentiation or the symbol that you're using is because of Jacobi.
- Jacobi actually reintroduced that symbol of Do, and then even today we use it.
- But there are other notations also for this.
- We will see how it is defined and, what it is and all.
- But the concept of partial differentiation was introduced by Leibniz, and then again it's like differentiating like it's like ordinary derivative.
- But you have to treat all other variables as constant and differentiate only the variable with respect to which we are finding the derivative.
- A partial derivative of a function of several variables is a derivative with respect to one of those variables with the other's held constant.
- And of course there are different notations.
- This is the most easier one for problems we preferably use this.
- You can find Zx, you can find ZY, and also you can find Z T.
- I am differentiating z like you differentiate any other function with respect to x, but with the constraint that you have to treat y and t as constant.
- As the definition is similar to the definition of the total derivative or derivative of a single variable function, all the properties, rules, and all those things that you used for the derivative of a function of single variable holds good here also.
- like product rule, quotient rule, chain rule, derivative of an implicit function, composite functions.
- All those things are applicable to partial derivatives also.
- I am taking bivariate, maybe I can do another one which is a multivariate function.
- Now, as soon as I ask you to differentiate, the first question that comes to your mind is with respect to what?
- I need to specify that.

Example or application from the lecture

- Suppose I have a function z is equal to f of x y.
- So let me take a simple example so that things will be a little more clear to you.
- Let's take a function which looks like say z is equal to.

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EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

don't see any other thing, no matter how many variables you had here.

Key details

- Or I can add an e to the power of W to this and an Ln of say some P.
- But when I ask you to find ZX,.
- All these things doesn't matter for me because all of them will become constant.

- And I just need to differentiate x cubed, which is 3x square, and you retain all these things as it is because they are constant.
- Let me choose another color so that you can visualize properly.
- What will be ZP now?
- What all will be constant?
- You don't need to worry about any of these functions.
- All these functions are constant.
- And you just need to differentiate Ln of P, which will be 1 by P and the remaining things will come as it is.
- The next thing is let me choose another color.
- The other terms will remain as it is.
- As I said, all the rules of differentiation hold good for partial derivative also.
- But I'll discuss it more when we do problems.
- Again, when I say all the rules of differentiation holds good for partial derivative, that we found the first derivative and then we extend it to find the second derivative, third derivative.
- You can continue that.
- But there is one thing that is different in partial derivative.
- And Zx might look like sine y into 3x square, something like that.
- And I'll call that as Z Y.
- This so what will what will I get?
- I have to treat 3x square as constant, but I can differentiate sin y which will be cos y 4.
- With respect to itself is higher order, that's what we saw now x, zx and z y.
- But I can differentiate zx with respect to y and then z y with respect to x.
- And they are called the mixed partial derivatives.
- What does this mean is first you go in the same order as we are prefixing.
- First you calculate the derivative of z with respect to x and then with respect to y.
- So if I'm doing z y x, then that means I will differentiate first with respect to y and then with respect to.
- And then another x.
- The first thing you need to do is find zx.
- Zxy you differentiate with respect to X again.
- And then you finally you differentiate with respect to X again.
- That is, if the order of differentiation is immaterial.

Example or application from the lecture

- Suppose I have say I have zx.
- I'll continue with the same example that I did in the previous page.
- So suppose let's take an example.

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EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

I would like to leave a note for you here to prove this Clydoids theorem.

Key details

- And the subscript notation, the proof is very simple.
- You don't need to do too much.
- Use the definition of limit definition of derivative and you can show that those two limits are equal.
- so next class we will see what is the geometry of this partial derivative.
- What geometrically what does the first derivative represent, second derivative represent.
- For a function of single variable, you already know that, right?
- The first derivative represents velocity, second derivative represents acceleration.
- But now we are in a higher dimension and we want to check what it represents.
- So that we'll take up in the next session.

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EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

3 Introduction to Partial Differentiation

Study focus: Introduction to Partial Differentiation. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000022 · EV0000023

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Anita from Department of Science and Humanities.

Key details

- As the name itself suggests partial differentiation is differentiating something partially.
- Maxima and minima for functions of two variables, Lagrange's method of undetermined multipliers, and as an application of all the Ebo concepts you will be learning errors and approximations.
- I know most of you might not like math that much, right?
- The only reason that you're here is maybe by force or you want to I hardly doubt that many of you want to learn things by choice, right?
- To make things a little more interesting, whenever we teach a topic, it's very essential that we ask ourselves questions.
- That is, why I need to study the topic, where exactly I'm going to apply the topic, and how can I apply the topic.
- How do we calculate the limit, continuity, and differentiability for a function of more than one variable?
- When we find the derivative, how can I apply that to find the relative extrema of functions?

- By the end of this unit, you will get the answers to all these questions.
- If we write like this, we know that y is a dependent variable which depends on x.
- But now we want to define a function which is a function of two independent variables.
- This is called a bivariate.
- Because both x and y are independent, we'll assign this function to another variable called z.
- whatever we do in this unit, we'll restrict ourselves to a bivariate function, but same thing can be extended to a multivariate function.
- The first case would be when you find the area of a rectangle.
- that area will be a function of just length and width.
- Electric potential if you want to find in within a cathode ratio, we know that it's a multivariable function with spatial coordinates as the variables as well as time t.
- The propagation of electromagnetic wave, which depends on the magnetic field, electric field, wavelength, and the direction of propagation.
- As you observe, this is a function of four independent variables.
- Not just that.
- you would need some parameters before you say what could be the weather in the next twenty four hours, right?
- But if there is a cyclone or any other major change in the weather, then you might have to include other parameters to decide the or to predict the weather conditions.

Example or application from the lecture

- On an larger scale, we can define a multivariable question as a function as quantities which depends on two or more variables are represented using functions of two or more variables, for example.
- Suppose you have a function of single variable.
- Let's say you have a function y is equal to f of x.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 05:00–09:59

the area, the population in that area, awareness among the people, the immunity level of the people who are living in that area, and the distribution of population based on the age group, gender, and other classifications.

Key details

- As you see, the Ebo examples make one thing clear that we are surrounded by function of more than one variable.
- A function of two variables denoted by so here we are denoting it as z is equal to f of x y.
- is a rule that assigns to each ordered pair of real numbers XY in a set D a unique real number denoted by f of x y.
- The set D is the domain of f and its range is the set of all the values that it takes.
- When do we say that this is a function you will have a domain which is a set of all real numbers.

- And then you have another set of real numbers called the range.
- And then we will map every element in the domain to a unique element in the range by this function f.
- But instead of having a real number here, I'll have an ordered pair here.
- When I put x is equal to 2 and y is equal to 1 in this expression, this will be mapped on to 8.
- The next question is, how does a function of more than one variable look like?
- It's just a parable which is symmetric about the y axis.
- But now if I ask you to plot a function z is equal to say sin x into cos y, it you can observe that it's not that easy for us to visualize how it looks.
- So we have luckily we have tools now which will help us to plot functions of n variables, at least two variables, because as I in increase the number of independent variables, the coordinate system has to increase.
- For a function with three independent variables, I have to add another dimension, which is practically not possible for us to demonstrate.
- Now this is the graph, the first graph that you see is for the function sin x into cos y.
- And this graph here is for a bivariate function again, which looks like e to the power of 1 minus x square minus y square.
- How does the graph of the multivariable function look like?
- The next thing would be we'll see how to find the limit of a function of two independent variables.
- Then we say that limit of f of x will go to a value L.

Example or application from the lecture

- But if you want to conclude or make a decision or draw a conclusion about any given situation, we have to consider always a function of more than one variable.
- What I mean by this in simple terms is suppose you have a function z is equal to f of x of single variable.
- For example, let's take suppose I have z is equal to x cube y.

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EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 10:00–14:59

As x approaches to A, if the only condition that I need to worry is as the distance between X and A reduces, the distance between f of x and l reduces.

Key details

- And then you calculate what happens to the value of the function.
- In both the cases, if it approaches to the value L, we say that the limit of the function exists and it is equal to L.
- We can do a similar thing for a function of two variables now.
- I say that the function f of x, comma y will tend to a limit L as.
- What do I mean by that?

- As I was mentioning, in case of single variable function, x will approach a either from the left or from the right.
- Let me choose a different color to show different path.
- It could be on a parabola.
- If we simplify this, we get 4 plus 4 minus 3 minus 8, this would be plus 3, and then minus 3 and then minus 6.
- I can cancel this, I can cancel this, and the final answer would be minus six.
- This is the graph of this function.
- How am I going to prove that it does not exist?

Example or application from the lecture

- In fact, what you did is suppose this is the line where I have my point A, then you can let X approach to A either from the left or from the right.
- X comma y approaches to a point, let's say x comma y approaches to the point, a comma b, if and only if this function f of x y takes the same value L.
- When xy approaches to a b along any given path.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 15:00–19:59

And that this line y is equal to 0 is nothing but x axis.

Key details

- In that case, what happens to the limit?
- I have to calculate limit as y is 0, I will get only x approaching to 0, 2x into 0 divided by 3x square plus 0.
- This will be anyway 0.
- What happens if it approaches along y is equal to x square?
- Wherever I have a y I need to replace it with x square, right?
- And if I simplify this, I'll get 2x cube divided by x square.
- If I take out x square common, I will get 3 plus x square.
- Of course I have limit.
- But what happens if I take the value that xy approaches to 00 along.
- The line y is equal to x.
- Wherever I have y, I'll replace it with x divided by 3x square plus x square.
- And if you observe carefully, this will be just 2x square divided by 4x square, and of course this is just half.
- That means the limit of this function does not exist.
- But it's fairly easier for us to check if the limit does not exist, because we can take two random paths and show that the value is not equal.
- We say that a function y is equal to f of x is continuous if it satisfies three properties.
- The first one is limit as extending to A of F of X must exist.
- The second thing would be f of a must exist.

- And the third one is limit as extending to A of F of X must be F of A.
- Similarly, in the next class we are going to define well how to define the continuity for a bivariate function or in general maybe for a multivariable function.
- So I'll see you all soon in the next class.

Example or application from the lecture

- Let's take along a different path.
- As you saw in the previous example, it won't be easy for us to check if the limit exists.
- Let's define now continuity for a function of a single variable.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

4 Geometrical interpretation of Partial derivatives

Study focus: Geometrical interpretation of Partial derivatives. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000038 · EV0000039

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We were discussing about the differentiability of a multivariable function and we call that as a partial derivatives.

Key details

- We also defined what is a higher order derivatives and what is the mixed order partial derivative.
- How to find ZX, how to find ZY, and if you had observed we actually did these many second order partial derivatives, right?
- But of course these were equal, but still.
- second order derivative, which are Zx, ZYY, ZXY, and ZYX.
- in M to the power of N order.
- That means you the first order derivative will have two terms.
- The second order will have.
- Again four terms for a two variable and so like that it continues.
- Then what is the dy by dx represents?
- that the derivative always gives me the change, right?
- But change in what?
- That's for a single variable function.
- I have a points which lie on the circle.
- You see a circle in the bottom?
- And that lie on this curve.

- It gives you the slope of the tangent.
- Something like this, and if you have a point, at this point, you please don't think that I am intersecting, this is a tangent because it's in 3D, it looks like that.

Example or application from the lecture

- A function of two independent variables, suppose you have a function of two independent variables, which is a bivariate function, it has two first order derivatives, which are zx and zy, and it has four derivative.
- That means, suppose I have a function f which is a function okay let me call z as a function of n independent variable.
- Suppose you have a function of m independent variable, then you will have m to the power of n derivatives of order n.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 05:00–09:59

So because it's a surface, at any given point you will have infinitely many tangents to the surface.

Key details

- Are you following what I'm trying to say?
- Let me erase all these things.
- There are infinitely many such tangent.
- I just want to find the slope of one such tangent.
- What is the tangent?
- That tangent is the variable with respect to which you are going to differentiate.
- Then Zx will give me, so I am calculating Z x a at X0 Y0.
- This derivative at this point will give me the slope of the tangent to the curve at x0 y naught, which is parallel to x axis.
- Are you following?
- The partial derivative with respect to y gives me the slope of the tangent.
- This gives me the slope of the tangent to this surface at this point.
- And what is the tangent?
- That tangent which is parallel to the y-axis.
- So the per first order partial derivative with respect to x gives me the slope of the tangent parallel to x axis and z y gives me the slope of the tangent parallel to the y axis.
- But the general question that would arise is.
- So that means what I if I had the my coordinate system like this and you had a surface.
- And the tangents that you got now is either it was parallel to x axis or it was parallel to y axis, something like that.
- But I want something the that is different.
- What how do I get the slope of this or what will this represent?
- This is ZX and this is Z1.
- I can resolve that into the component along x axis and the component along y axis.

- And this vector also gives me the slope of the tangent and this is called the gradient.

Example or application from the lecture

- Let's say that this is the surface which represents Z is equal to F of XY.
- So geometally the first order partial derivatives, suppose let me write this as zx and this is zy.
- Let's see how does it look like.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 10:00–14:59

As that like the first order derivative in the for one variable gives you velocity.

Key details

- And then D square y by dx square gives you acceleration.
- Do what do they call for the higher order derivative?
- T cube y by dx cube or d to the power of four y by d x power four.
- When you did maxima minima.
- It's not that easy for me to plot in 2D, but I'll show you a video.
- And then if f y is greater than zero, then it is concave up in the y direction.
- Same thing with the mixed order partial derivative.
- Fxy will tell me how the partial derivative of f x is changing in y direction.
- What is the change in f y along the x direction?
- I mean it's again, it's not that easy for you to visualize.
- This is just from Ultra Am lab.
- what I have done is I have animated that so that you can visualize for different values of fxx.
- So I have taken the cases where for the first case as you can see I have taken f of xx to be sorry negative.
- So in the first case I have taken FXX to be negative and then FYY also to be negative and then I have taken fx y to be positive.
- And then I have increased the value of x.
- And you can see that it is concave up along the y axis or x axis, sorry.
- And then I have changed now fx is positive, but I have changed x fxy to be negative.
- Yeah, it's it's nice to see these visuals.
- you can try by yourself by changing different values.
- just to check what happens.
- and this negative and this positive.
- And you can keep on changing the value of X and changing the value of Y.
- That will help you to visualize the tangent.
- Do you see the tangent here?
- , so this these things will be better if you try by yourself.
- in the next class what we are going to do is we'll try and solve some problems on partial derivatives.

- We'll try to find mixed order partial derivatives, we'll try to verify the Claret theorem and all those things.

Example or application from the lecture

- The second derivative so suppose what I mean is suppose you have y is equal to f of x, then dy by dx gives you velocity.
- But you also know that suppose you want to find whether the curve is concave up or concave down, you use usually the second derivative, right?
- Suppose I have the second order partial derivative which is fxx and if this is greater than zero, that means the function is concave up in the x direction.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 15:00–19:59

I want you to explore.

Key details

- I'll give you a link in the end of the class wherein you can try all these things.
- You can try how to plot a three dimensional graph, how to do this animation, and, how to check the partial derivative and his geometrical interpretation and other things.

Example or application from the lecture

- But I have given a link in my next slide.

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EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

5 Problems on Total derivatives, Chain rule

Study focus: Problems on Total derivatives, Chain rule. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000054 · EV0000055

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will solve problems on total derivative, chain rule, and implicit function.

Key details

- Point du by dt at a point t equals to pi by 2 where u equals x square y times a power x y and x equals t square plus 1 and y equals sine t.

- From the data, we can see that u is a function of x and y, where x and y are the functions of p.
- Now because we want du by dt at the point t equals to pi by 2, we need to find.
- X and Y at the point.
- At t equals to pi by 2, x becomes pi square by 4 plus 1 and y becomes 1 because of sine t.
- Also, du by dt becomes because y is 1, it becomes only e power x times 2x plus x square.
- We also see that when t is pi by 2, the second term doesn't exist because d by dt is cost t and the cost t is 0 at t equals to pi by 2.
- Pi square by four plus one of two times pi square by four plus one plus pi square by four plus one multiply.
- This is the value of d u by dt at the point d equals to pi by two.
- Find du by dt.
- We use a same approach what we have used for the previous problem.
- Plus two y divided by y square plus one.
- times second square t minus x e power x y plus two y divided by y square plus one into sin t.
- Now because u is a function of t, we express x and y in terms of t.
- The base of a triangle is increasing at a rate of 2 centimeter per second, and its height is increasing at a rate of 1 centimeter per second.
- At a certain instant, the base is 10 centimeters, the height is 6 centimeters.

Example or application from the lecture

- Let's begin with total derivatives.
- For the given x and y, d x by dt as well as dy by dt becomes 2t as well as cos t respectively.
- Let us look at a similar problem.

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EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 05:00–09:59

Find the rate at which the area of a triangle is increasing at that instant.

Key details

- And the height of a triangle is increasing at a rate of 1 centimeter per second.
- Also, p and h are respectively 10 and 6.
- It means the rate at which the area of the triangle is increasing is 11 square centimeters per second.
- Water is being poured into a conical tank at a rate of 9 cubic centimeter per second.
- Find the rate at which the water level is rising when the height is 6 centimeter.
- pi r square h.
- Because height and radius are equal when we replace r equals h, v becomes 1 by 3 pi h power 3.
- After calcation 3 and 3, we get dv by dt to be pi h square dh by dt.
- P V by D T is 9 and H is 6.
- It means u is a function of x and y, where x and y are the functions of s and t.

- then u becomes composite function of s and t.
- Do x by dou s is going to be t times e power s t. dou x by dou t becomes s times e power s t. becomes two s dou y by dou t becomes two t.
- Then find a value of douz by doux as well as douz by dou y at the point zero.
- We know from theory the formula for dou z by doux is minus of f x divided by f z.
- And similarly, dou z by dou y equals minus f y divided by f z.
- It means we take this as some function of xy and z, then we find only partial derivative of f with respect to x y.

Example or application from the lecture

- In the problem, it is given that the base of a triangle is increasing at a rate of 2 centimeters per second.
- Let us look at one more application problem on total derivative.
- It is given that the height and radius of the cone are equal.

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EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 10:00–14:59

as well as z.

Key details

- Then using the formula we can compute douz by doux and dou z by dou y at the point zero.
- Of course there are two ways to solve this problem.
- Another method what we can use we can use dy by dx equals.
- Minus of partial derivative of f with respect to x divided by f with respect to y.
- It means we can take this as some function of x and y, then use the partial derivative and use this formula, we can get the answer.
- So another method which is called direct method, which is shown as follows.
- Data x square plus xy plus y square equal to 7.
- Differentiate both the sets with respect to x.

Example or application from the lecture

- Let us solve another problem on implicit function with two variables.
- If the relation between x and y is given by x square plus x pi plus y square equals 7.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

6 Total derivative, Chain rule and Implicit functions

Study focus: Total derivative, Chain rule and Implicit functions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000066 · EV0000067

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Let's welcome you all to another session of partial differentiation.

Key details

- Today we'll be learning chain rule, composite functions, and total derivative of implicit function.
- It's like an extension of what you already did in the case of a function of single variable.
- There are three functions which you need to take it.
- The first one is exponential.
- Next you have to differentiate sine, which will be cos of x square.
- Next you differentiate x square, which will be 2x.
- So that means you had f of g of you had f of g of h of x.
- And when you differentiate it, you followed the order of differentiating f first, then g, and then in the end with respect to h.
- And then if x and y again are functions of one some independent variable, in this case it could be t.
- If you substitute y as 2t square, you will get 2t square, the whole q.
- But always it might not be that easier for me to substitute the independent variables and reduce it to a single variable and differentiate.
- So what we can do is we have another alternate method.
- I had z is equal to x square y cube.
- I took x is equal to t and then I took y is equal to some 2t square.
- What happens if I differentiate x with respect to t, I can get the total derivative because it's a function of single variable.
- I already know the partial derivative of z with respect to x and y.
- Plus dou z by dou y, which will be x square, I am keeping it as it is.
- You differentiate y cube with respect to y.
- T to the power of six.
- I think I took t square if I'm not wrong.

Example or application from the lecture

- For example, suppose I asked you to differentiate this function, which is y is equal to e to the power of sine of x square.
- This was just an example of a function which is a composition of two more functions because it's of single variable, right?

- For example, suppose I have a function z, which is a function of two independent variables x and y.

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EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 05:00–09:59

You can substitute x square again, so this will be t power 4, 3 into 2 t square, the whole square into 4 t square.

Key details

- And we can simplify this.
- So this is an alternate method of.
- Finding the total derivative of a function of two or more independent variables when each of the independent variables is a function of another variable.
- So that means whenever z is a function of xy.
- And x and y are functions of some parameter say t.
- In that case, I can write.
- I can actually find the total derivative of z with respect to a single variable t.
- I can do that by finding the partial derivative of z with respect to x into the total derivative of x with respect to whatever the variable you have t.
- Plus, I can also do z y dy by d.
- You can apply this formula when the independent variables are functions of a single variable.
- A paraboloid you can imagine like a cup, okay?
- With the base, of course.
- Something like that.
- So a paraboloid and the surface whose Cartesian equation that means I have a cylinder and a paraboloid.
- And then I have a surface whose Cartesian form is cylinder.
- I need to calculate how fast Z is changing when X is equal to two.
- I don't know what is dy by dt is, so I'm not worried about that.
- I need to calculate what is the value of z.
- Do I know z already?
- Yes, z is x square by 16 minus y square by 9.
- with respect to x will be x by eight.
- Partial derivative of z with respect to y will be minus 2y by 9.
- that this is dou z by dou x as z is a function of xy into t x by dt plus.
- Do z by dou y into t y by d.
- This is what I have.
- We can substitute x is equal to 2, that's not a problem.
- I can find this, I already know this, I know this, I know this, but I don't know this.
- So let me erase that first.
- that this is an implicit function, right?

- If y is a function of x, so I can definitely differentiate this, I'll get 2x plus 2y into dy by dx actually equal to 0.

Example or application from the lecture

- So single variable, for example.
- Suppose I have a I have a problem now, okay?
- A point P is moving along the curve of intersection whose Cartesian equation is given.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 10:00–14:59

Can I do this?

Key details

- I'll get dy by dx equal to minus x by y.
- But I don't want dy by dx.
- What I want is dy by dt.
- These things are known to you when you did the first order partial derivatives.
- Minus x by y.
- You can also find what is the value of y when x is equal to 2.
- when x is equal to 2.
- And what is the formula for dz by dt now?
- You have zx into dx by dt.
- plus ZY into TY by D.
- Zx we got it as x by 8 into dx by dt.
- Z y also we know which is minus 2y by 9.
- dy by dt, I just now got that to be minus x by y into dx by dt which is again 0.2.
- Just put x is equal to 2, you will get the answer.
- If dz by dt is negative you can say that z is decreasing at say x is equal to 2.
- At x is equal to 2, I'll get 0.2 divided by 4, and then plus or minus or plus.
- This will become plus and x is again 2, I have to substitute.
- All I'm interested in to know the behavior of the function.
- They want you to calculate the value of the function at a different point.
- So that means I x is there is a small change in x which is 0.1 and a small change in y which is 0 point.
- Again, you just apply the definition of dz.
- If I don't want to restrict myself to any particular variable, I can write the equation like this also.
- And you substitute the values that are there and you can find that approximately it is equal to 6.418.
- I have another problem.
- If the area of a triangle remains constant,.

- So what is at what rate does the angle between the side should change when the first side is 20 centimeter and the second is 30 centimeter.

Example or application from the lecture

- But first, let's find out what is dy by dt is.
- Suppose now what is required is dz by dt, correct?
- They have already given it as 0.2.

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EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 15:00–19:59

They have given me this side and this side.

Key details

- I want to know at what rate this theta should change, whether theta should increase or theta should decrease, so that the area of the triangle remains constant.
- And the area is half into base into height.
- sine theta will be opposite side by hypotenius.
- That's how I got this expression half x y sin theta.
- So, because there are three independent variables, you calculate the partial derivative with respect to x, dx by dt, with respect to y, dy by dt, with respect to theta, d theta by t.
- And what is required is this, they say that area remains constant.
- So what is the meaning of that area remains constant means dA by dt is 0.
- I don't want any change to happen to this area.
- Again, I have another wherein you have to conclude certain assertions.
- And four feet across the diameter.
- It is known that the steel will expand or contract with temperature changes, right?
- or in the height of the tank.
- You need to tell me whether the change in the diameter or change in height will affect the volume.
- This is also simple.
- And we conclude that dv is equal to 40 pi dr plus 4 pi dh.
- If you observe carefully, so I got an expression that the volume is changing as 40 pi dr plus 4 pi dh.
- Because the coefficient of r is 10, definitely the radius would contribute more towards the change in the volume than the height.
- Just by doing a little bit of differentiation, you can conclude what is the change that is happening, what is contributing to the maximum change.
- And then each of them were function of t.
- This is what I had in the previous case.
- This is what I did.

Example or application from the lecture

- Suppose I'll draw a perpendicular because I need the height.
- Let me take the base as x and suppose this side is y, then that sine theta, right?
- So I hope you appreciate this example so that things will be that much clearer to you.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 20:00–24:59

Let's change little bit.

Key details

- Z is a function of two independent variables, say x as well as y.
- But both x and y are functions of again two independent variables u and v.
- That means what I have to differentiate Z partially.
- But Z is a function of XY.
- Z u will be Zx into XU plus.
- Z y or ZY into Y U.
- I need to calculate this Z U, so I'll calculate Z with respect to X and then with respect to Y and then I'll go like this.
- It can be a function of n variables.
- Then you can find the partial derivative of z with respect to all these three independent variables.
- But you don't need to restrict to only two independent variables, you can move on and find out what is it for a multivariable function as well.
- Then I need to prove that 6vx plus 4v plus 3vz is 0.
- And each of them R is a function of XY, S is a function of X and Y, and T is also a function of X and Y.
- That means I can calculate what is Vx.
- What will be Vx?
- Vx would be VR Rx V S X V T X.
- And it's easy for you to calculate them.
- S X is just minus 4 and S y is just 3.
- And T we have taken it as 4z minus 2x.
- Let me use an eraser.
- this is ZX.
- How do we get Vx?
- Vx is VR R X.
- Then again V S, there is no X there, but you have to write.
- And then V T X.
- So this term here is 0 because I don't have s as a function of x.

Example or application from the lecture

- Let's take an example now.
- They have given V to be a function of this.

- Let's just move a little bit.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 25:00–29:59

I have another example.

Key details

- It's always better if you write the tree diagram every time you have a function of more than one independent variable, so that you won't do any mistake in calculating the partial derivative.
- R is a function of T and S and theta is also a function of T and S.
- I want Zs.
- I have to keep it as it is because I don't know what the function is.
- See if here what will be Rs, Rs will be just T and R T will be S minus 2T.
- And z theta, I don't know, I have to keep it as it is, as I don't know what the function is.
- But theta s of course I know.
- 2 root of s square plus t square into 2 s.
- I can get rid of this.
- But the problem with this problem is you don't know what is this function.
- I need to retain them as it is in the solution as well.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

7 Implicit functions and problems

Study focus: Implicit functions and problems. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000090 · EV0000091

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

It will be a continuation of chain rule and composite functions and other things that we saw in the previous video.

Key details

- which is like f of x y equal to c some constant, then we call this as an implicit function.
- This is an implicit function.
- I have x cube y plus 3xy square equal to 3.
- How do we differentiate this?

- You differentiate this using product rule, keep the first function as it is.
- Derivative of y is dy by dx, plus, keep the second function as it is, and you differentiate the first, which is 3x square, plus.
- You keep the first function as it is, you differentiate the second again, 2y dy by dx plus I'll keep the second function as it is and I differentiate.
- The advantage of this is the right hand side will become 0.
- Because it's a constant.
- And from this expression, it's easy for me to find what is dy by tx.
- or f of x y equal to some constant.
- But now if z is a function of xy, but with a restriction.
- That restriction you have to remember.
- That z must be a constant.
- So here if z is a function of n independent variable and if it's equal to constant, I can always find the total derivative of z which is defined by.
- DZ is equal to, so this, dz is equal to partial derivative with respect to x, dx1 with respect to just now we discussed what is the total derivative, right?
- If z becomes constant, then it looks like f of x comma y equal to c.
- This is called a level surface.
- And it's not new to you, right?
- Or isotherms where at every point the temperature remains the same.
- that this is nothing but a sphere centered at the origin, right?
- If C1 become 1, I'll get the unit sphere.
- C1 become 89, I'll get a bigger sphere.
- Every color would represent a different value of C, and it shows that for every color the temperature distribution or whatever you want to call it, the pressure or the potential or distribution of current, everything was remains constant.
- Of course, it's difficult to plot because I don't have that many dimensions to plot, but definitely it would represent a level surface.
- And in case of level surface, you can find the total derivative.

Example or application from the lecture

- For example, you can see here I have an problem x cube y plus 3xy square equal to 3.
- And there is a special name given to this.
- The level surface, for example, you take equipotential surfaces.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 05:00–09:59

Let me go back to my previous slide so that it will be easier for you.

Key details

- But is there another way that I can somehow make use of a partial derivative to this?
- If you observe carefully, you have this, right?

- Let me take this as some set.
- Isn't it ZX 3x square y plus 3y square?
- I am differentiating z partially with respect to x.
- And what is z y?
- Z y will be.
- Again I get x q plus.
- 6x y and if you observe carefully I got this dy by dx as minus zx by sec y.
- This will be useful if your function is too complicated.
- the see the tedious task that we did here I have to use product rule maybe quotient rule or chain rule or whatever it is the function might be as complicated as possible but it's not possible for me to find the d by dx very easily or very conveniently.
- But now if I need to find the total derivative, you already know z is a constant, so that's not going to change.
- That was a small observation.
- And that was a good observation.
- And this df is called the differential.
- In fact, you have already used this in your course.
- And where did we use this?
- How did you do this integration?
- As soon as you look at the integrand, it's clear that you have to make a substitution.
- That is where I can apply the total differential.
- That's mean basically I'm I'm in x plane, now I'll be in dt plane.
- All my x square x gets transformed and I'll get only a simple function as z to the power of t which you can integrate directly.
- Another appro application would be in errors and approximations that I'll discuss by the end of this unit when we do applications.
- These two are independent.
- those things you can find.

Example or application from the lecture

- I differentiate it given expression with respect to X and Y, and then this is the expression I got.
- Can you find out what is ZX from the given expression?
- And again, I need to consider that z is a function of xy, otherwise it's difficult for me to express it.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 10:00–14:59

Z y so this is like an extension of what we just discussed to another variable.

Key details

- One more variable is added.

- Again it represents a level surface.
- what I want you to concentrate is Zx.
- maybe I'll give you x cube y square plus three x y z plus some sine of e to the power of z this is equal to some c just imagine how much effort you need to put to differentiate this with respect to.
- X partially, right?
- You have to separate Z and then or because X and Y are both independent.
- You'll get you don't know how what to do with Y X and keep treat Y as a constant and retain that and everything.
- But here this formula will tell you that dou z by dou x is minus f x by f z.
- You can extend it to n number of variables.
- So the first question is for the curve x e power y plus y e power x equal to 0, find the equivalent of the tangent line.
- That means it is of the form z is equal to some function of xy or y x equal to zero.
- That's from the definition of total derivative.
- F of x comma y is x into e to the power of y plus y into e to the power of x.
- That's f x.
- F y will be x into e to the power of y plus e to the power of x.
- divided by X e to the power of Y plus e to the power of X.
- This term will vanish.
- The question says, find dy by dx if cos x par y is equal to sine y to the power of x.

Example or application from the lecture

- Suppose you have f of x y z equal to c, then zx is minus f x by f z.
- I know that the total derivative for this is given by so in this case if f of x y is equal to 0, then dy by dx is nothing but minus f x divided by f.
- If you need to find the tangent in any other point, let's say I want to find at the point x0 y naught, then you can write this as y minus y naught is equal to m into x minus x naught.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 15:00–19:59

Can we rewrite this?

Key details

- That means this represents an implicit function for me now.
- See, just imagine if you differentiate this directly, you need to differentiate like first cos x and then apply chain rule to this, it's to the power of y, and then sine y to the power of x.
- Things will become more complicated, right?
- But if we use partial derivatives, things will be much simpler.
- The formula for dy by dx will be minus fx divided by f y.
- When I differentiate this with respect to x, y is a constant.
- Is this clear what I did?

- So I had this as the first.
- First this is just the first term.
- Ah derivative of cos x is minus sin x.
- What is the derivative of a power x for you?
- If you remember derivative of a power x will be a to the power of x into ln of a.
- That's my numerator.
- That's the numerator, okay?
- So that means cos x will become a constant for me now.
- And derivative of a power y will be cos x to the power of y into ln of cos x.
- Minus same thing I have to do.
- This is a variable for me, but x is a constant.
- Of course, not much simplific simplification can be done for this, but this is the expression that you get for dy by dx.
- But you need to be careful.
- that x cannot take the value 0, neither y can take the value 0 because ln of 0 is not defined.
- So we'll try the next problem, which is again on implicit functions and we'll see how it goes.
- The question is find dou y by dou x and double at the point one comma zero comma two for this particular function.
- That's what is expected.
- And when I differentiate z partially with respect to x, I need to treat y as constant.
- derivative of y with respect to x by taking f minus of f x by f y.
- This is using the formula for total derivative.
- Where what is f?
- F is Z cube plus XY minus Y square Z.
- Please Z is constant.

Example or application from the lecture

- So let's differentiate the exponent first, which is y cos x to the power of y minus one.
- If any particular point is given, then we may continue and substitute that point and find out.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 20:00–24:59

I have two terms with y in it.

Key details

- Minus two Y Z.
- This is what I get.
- But I need to calculate y x at the point 1 comma 0 comma 2.
- At the point 1 comma 0 comet the value of y, y is 0 divided by x, x is 1 minus 2 into y is again 0 and z is 2.
- So that means I'll get this value as zero because the numerator is zero for me.
- Dose by dou x will be minus f x divided by f z.

- There's only one y term and f z will be three z square minus y square.
- I am differentiating f partial b with respect to z.
- But again I need to calculate this at the point.
- One comma, zero, comma two.
- So because y is zero, this is also zero for me.
- Tangent of the slope or the slope of the tangent at this point parallel to x axis is a horizontal line and even at this point it's a horizontal line.
- In our next session we will try some problems on Taylor's and McLaren series.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

8 Problems on Composite and Implicit functions

Study focus: Problems on Composite and Implicit functions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000110 · EV0000111

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will solve problems on composite functions.

Key details

- Then find the value of do u by dou t.
- It means u becomes the composite function of r and t.
- dou u by dou t equals dou u by dou x into dou x by dou t plus dou u by dou y into dou y by dou t.
- Then dou u by dou t becomes 2x into cos of x square plus y square into minus r sin t plus 2y cos of x square plus y square into r cos t.
- Now because u is a composite function of r and t, we eliminate.
- Let is x and y.
- R sin t into cos c.
- Absolutely problem here.
- U is a function of x and y.
- X and Y are the functions of R and T.
- Find dou u by dou theta where u equals x square y plus y z q, x equal to r cos theta, y equal to r sin theta, and z d equals to theta.
- In this problem, u is a function of x y z.
- where xy z are the functions of r and theta it means u becomes the composite function of r and theta so in this problem we want only one particular case that is dou u by dou theta so when we use chain rule.
- W by dou y is x square plus z power three.

- W u by dou z equals three y z square.

Example or application from the lecture

- Let us look at the first problem.
- It is given that u equals sin of x square plus y square where x equal to r cos t and y equal to r sin t.
- It is given that u is a function of x and y, where x and y are the functions of r and t.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 05:00–09:59

[ASR low confidence; check audio].

Key details

- If u equal to a function of r where r equal to square root of x square plus y square plus z square, then express dou u by dou x in terms of f dash of r.
- When r is a function of x, y, and z.
- Because r is square root of x square plus y square plus z square, so the partial derivative of r with respect to x is going to be x divided by square root of x square plus y square plus square is after calculation up to 12.
- And x and y are the functions of r and theta, then u becomes the composite function of r and theta.
- So what would be the next step in this particular problem?
- Can you stop here?
- Because u is a composite function of r and theta, we need to eliminate this x and y.
- Zero because after cancellation of this and this.
- If u equal to tan in us y by x, when x and y are the functions of r and theta, then show that 2 u by 2 theta equals 1.
- Again after writing the particular chain.
- Then when we compute dou u by do x, dou u by dou y, then dou x by dou theta and dou y by dou theta and finally upon using x and y to be r cos and r sign respectively.
- We see that two by dou theta becomes one.
- U is a function of x and y.
- When x and y are the functions of s and t then u becomes the composite function of s and t.

Example or application from the lecture

- Let us look at this particular problem.
- Again, when we look at the chain, it is given that u is a function of r.
- So when we write the chain rule for dou u by dou x, it be it becomes dou u by dou x equals dou by dou r into dou r by dou x so because u is f of r so dou u by dou r becomes f dash of r into dou r by dou x so let us find the value of dou r by dou x.

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EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

Core explanation · 10:00–14:59

And you must know why did we use x equal to e power s and y equal to s plus t after this particular step.

Key details

- So because u is a composite function of s and t.
- Then find the value of dou by dou x and dou by dou y.
- U is a function of r and theta where r and theta functions of x and y.
- Then u becomes composite function of x and y.
- Then when we use chain rule, we get dou u by dou x has this and dou u by dou y has this so after solving for dou r by dou x dou dita by dou x dou y and dou dita by dou y we get this and this as the answer for this particular problem.

Example or application from the lecture

- It is given that u equals f of r theta, where r equals square root of x square plus y square and theta equals that inverse y by x.

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EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

9 Homogeneous functions and Euler's theorem

Study focus: Homogeneous functions and Euler's theorem. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000122 · EV0000123

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we'll be learning something called as Euler's theorem, which is a wonderful observation made by Euler for a certain class of functions.

Key details

- The advantage and what are the uses for this Euler's theorem we can discuss later.
- Before we go to Euler's theorem, we need to refresh our memory about homogeneous functions.
- you might have heard of homogeneous function when you solved differential equations.
- And that was the most easiest one because any homogeneous function you could reduce into a variable separable form, right?

- I hope you remember this, right?
- So that means I was in a comfortable position to reduce this to variable separable by making the substitution y is equal to bx.
- But now what happens is we will discuss.
- We will see how to check a homogeneous function.
- I have a function x square plus y square.
- Or I may take a function which looks like x cube plus three x square y plus y cube plus three x y square.
- I know that this is homogeneous, but I'm not sure whether e whether e po of that is homogeneous.
- I don't know how to check this function to be homogeneous.
- If at all it is homogeneous, what is the degree of that function?
- One is the function homogeneous?
- And if it is homogeneous, what is its degree?
- Again, I'm restricting myself to a bivariate in the beginning.
- But if you want to go for a multivariable function, definitely you can do it by extending the definition.
- If it can be expressed in the form of either x to the power of n into some function of y by x, or if I can write it as y to the power of some m into some other function of x by y.
- Then we say that the function is homogeneous of degree n in this case and m in this case.
- What do I mean by that?
- How do I check if this is homogeneous?
- If I take out x square common, can I write this function as 1 plus y square pi x square.
- That means this function is homogeneous of degree.
- A function f of x y is said to be homogeneous of degree n.
- If it can be expressed in the form of x to the power of n into function of y by x or y par n into function of x by y.
- Whichever is convenient, you can choose according to the problem.
- Then how am I going to find if it is homogeneous or not?
- Again we'll use the same definition, but now it's not just y by x, but it is z by x also.
- you have x q plus y q plus z q plus three x y z I can take out x cube common and if I take out x cube common I can g get the remainder remaining terms in functions of y by x as well as z by x.

Example or application from the lecture

- But let me give you another example.
- Suppose I'll write a function which looks like sine inverse of x plus y plus z divided by something like x cube plus y cube plus z cube, or it could be tan inverse of something or e to the power of something.
- Let's say the definition says that a function u is equal to f of x y.

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EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 05:00–09:59

Or it could be other way also.

Key details

- It could be using z power n.
- Z power n into function of x by z and y by z.
- We call that is a homogeneous function of degree n negative.
- Or you can write this as y to the power of n into some other function of now x by y and z by y and w by y and that continues.
- So before I go to Euler's theorem,, by the way, he is Euler, as you can see in all his photographs, his one eye is shorter because he was blind by one eye.
- , in spite of that, he was an extraordinarily intelligent mathematician.
- He didn't had any former schooling or anything, but of course he was with the family of Legrange or Barnooli, I don't remember, but.
- That's how if you're interested you can read more about who Euler was and what all his contributions are.
- As you can see, a constant function definitely is a homogeneous function of degree zero.
- And of course the converse also works.
- And it need not be just positive integer.
- Degree can be positive, negative, zero, could be anything.
- And I can take out an x common from the denominator, right?
- This is 1 plus y by x.
- That means this function here is a homogeneous function of degree minus half.
- It's not always positive, negative.
- It could be rational, it could be positive, negative integer, or even zero.
- It's okay.
- Ah so the remaining results I'll come back to these things when I define Euler's theorem.
- This is the most important one.
- Then when I multiply x with the partial derivative of u with respect to x, y with the partial derivative of u with respect to y, it is same as n multiplied by the function itself.
- That's what I said.
- Check the degree, then you can say directly that x u x plus y u y.
- No wow, what is it happening geometrically?
- I'm finding the slope of the tangent parallel to x axis, but I'm multiplying that with x.
- I'm finding the slope or change along the y direction multiplied by y.

Example or application from the lecture

- Let's say you had another variable w, it should be w by x.
- And linear functions as we saw in the previous example, so we took 3x plus 4y and we saw that this linear function is of degree 1 always.
- So just to show you an example, suppose I had say root x plus root y divided by x plus y, if suppose this was my function, that I can take out root x common from the numerator and write this as 1 plus root of y by x.

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EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 10:00–14:59

There is a change that is happening along the y direction for a surface, right?

Key details

- You study all those changes, then what you get is stacking of the functions one above the other end times.
- That's what you get.
- Ux is 2x and uy will be 2y again.
- Plus 2 y square, or this is just 2 into the function itself.
- And this is a homogeneous function of degree 2.
- So basically what Euler observed is for a homogeneous function, homogeneous means which is smooth everywhere.
- For a homogeneous function,.
- The change along the x direction when scaled with x.
- Change along the y direction scaled with y will give you the entire function itself scaled by the factor of the degree.
- This is for the first derivative.
- We discussed this in the previous lectures.
- You have ux y and uyx.
- But definitely they are equal.
- Okay, because first order partial derivatives are continuous.
- So what Euler says is because this is second order, you scale it by 2.
- How many third order partial derivatives can exist?
- It is U XXX, right?
- Uy and then U XY.
- U X Y X.
- And then U X Y.
- And then U Y X and then U Y X Y.
- It's of order three, right?
- No x y x I did.
- X Y is also done.
- But anyway, so you can check the mixed order partial derivative and what happens to the right hand side will be n into n minus one into n minus two into u.
- That's what you're going to get.
- As in when you add.
- one more variable to it.
- Then if it is homogeneous, so u is equal to f of x y z is a homogeneous function of degree n, then I get x u x plus y u y plus z u z equal to u.
- At this point I want you to explore.
- What is the corollary one that we did?

Example or application from the lecture

- but let's see what is the extension of Euler's theorem.
- Suppose because I have only ux and u y.

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EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

Core explanation · 15:00–19:59

and see what is the least recent research that is happening in this field.

Key details

- As I was telling, basically Euler's theorem is to relate homogeneous function with their partial derivative.
- See, that's what Euler observed.
- But if I have a partial derivatives of a function of two independent variables, how am I going to relate them with the degree of their homogeneity?
- and then it has extensive applications in economics.
- Actual form of oiless theorem they use in production function, marginal productivity, elasticity of demand.
- I didn't put the equations in detail because,, most of you are engineers and you might want an engineering application to Euless theorem.
- And intensive variables are homogeneous functions of degree.
- Maybe in your physics course you can mm better understand what is happening and what is the Euler's theorem.
- what does it say and other things, right?
- Because the Euler's theorem is the most important one wherein you can reduce the problem.
- Actually you don't need to find the partial derivatives at all, right?
- The first thing that we did is we defined a homogeneous function.
- We defined what's a homogeneous function.
- Basically, along with the degree.
- And if the degree of the homogeneous function, then Euler's theorem says that x into ux plus y into u y is equal to n into u.
- And then we also observe the extension of this for the second order.
- N independent variables.
- So in the next session we will solve problems on Euler's theorem.

Example or application from the lecture

- So basically it also gives the scaling of the given function, that's because of n into u, that's what we discussed just now.
- So here I have taken three independent variables and I have given first order.

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EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

10 Problems on Euler's Theorem

Study focus: Problems on Euler's Theorem. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000138 · EV0000139

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Solve the problem let's check the definition of Euler's theorem.

Key details

- If f of x y is homogeneous of degree n, then x fx plus y f y is equal to n into u.
- And it's we can check that this function is homogeneous conveniently by taking an x common and root x common from the denominator.
- That's a statement of Euler's theorem.
- The partial derivative of u with respect to x.
- Ux will be you can apply either quotient rule or product rule.
- That's u x.
- And similarly we can write u y to be root x plus root y minus x plus y into 1 over 2 root y'all.
- whole divided by root x plus root y the whole square.
- Plus Y, U Y will look like.
- So if I multiply the first equation with x, I can write this as x by root x plus root y after some simplification, right?
- I can cancel this minus I'm multiply it with x.
- x plus y into root x divided by root x plus root y the whole square.
- This is what I get from the first equation.
- I can go with the second equation which is y into u y.
- into root y for the second term.
- So if you observe carefully I can group these two terms because they have common denominator and I can group these two terms.
- So if I group these two I'll get x plus y divided by root x plus root y that's for the first term minus 1 half into if I take out x plus y by root x plus root y the whole square I will be left with root x in the first case plus root y in the second case.
- And that will give me half of x plus y by root x plus root y.
- And I know that this is nothing but u.
- Started with the left hand side.
- X UX plus Y Uy and whatever the resultant I got is the right hand.
- To apply Euler's theorem it's difficult.
- See just imagine differentiating this with respect to X, with respect to Y, with respect to Z, substituting here.

Example or application from the lecture

- Let's try another problem on Euler's theorem.
- Instead what we can do is we can take let's take uppercase z to be some sine u.

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EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 05:00–09:59

which will be x plus two y plus three z divided by square root of x par eight plus y par eight plus z to the power of eight.

Key details

- I don't need to worry about what is here because I can write this as x to the power of minus 3 into some function of y by x, comma z by x.
- Which is nothing but Z, this uppercase Z.
- That means z is a homogeneous function of degree minus three.
- That means we can apply Euler's theorem to Z, which says that x into zx plus y into z y is equal to minus 3 into z.
- That means ZX will be cos u into u x.
- and Z will be cos u into U Z.
- If we substitute these things in this equation we will get x Z into cos u into U Z plus X into.
- Cos u into u x plus y into cos u into u y.
- This is nothing but minus 3 into z.
- Z is nothing but sine u.
- So I will write x first.
- So I will get x u x plus y u y plus z u z is equal to minus three tan u because I divide it through out by cos u I'll get this result which is the required result.
- We could just check if the function is homogeneous, apply the statement of Euler's theorem, and we establish the result by taking sine.
- Next we will solve some problems approximation.

Example or application from the lecture

- And let's make a small rearrangement of this.

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EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

11 Problems on Euler's theorem

Study focus: Problems on Euler's theorem. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000146 · EV0000147

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will solve problems on homogeneous functions and Euler's theorem.

Key details

- Remember, here n can be positive, negative.
- Or even zero.
- Then we are left out with x power 4 plus 2x square y square plus y power 4.
- In the denominator we get lambda power 4 of x square plus y square whole square.
- After lambda power 4, lambda power 4, we can write 1 to be lambda power 0.
- U equals sine inverse of y by x divided by square root of x square plus y square.
- Replace x by lambda x, replace y by lambda y.
- Sign in us of lambda y by lambda x.
- And in the denominate denominator, it is square root of lambda x whole square plus lambda y whole square.
- U equals x square y plus xy square divided by x plus y power 3.
- U equals e power x minus e power minus y divided by x minus y.
- This simplification doesn't give us lambda power n times u of x comma y.
- It means u of lambda x comma lambda y is not equal to lambda power n times u of x comma y.

Example or application from the lecture

- Let us solve the first problem u equals x power four plus two times x square into y square plus y power four divided by x square plus y square one square.
- Now in order to test whether the given function is homogeneous or not, we need to prove this particular condition that is u of lambda x comma lambda y equals lambda power n times u of x comma y.
- In the given function, we replace x by lambda x, y by lambda y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 05:00–09:59

I hope the concept of homogeneous function is clear to you.

Key details

- If u equal to x minus y divided by x square plus y square, then find the value of x into dou u by dou x.
- plus y into dou u by dou y and x square into dou square u by dou x square plus 2 x y into dou square u by dou x into dou y plus y square into dou square u by dou y square.
- As we know that this is nothing but extension of Euler's theorem.
- So in order to use Euler's theorem, first we need to prove that u is a homogeneous function of degree n.
- We get to know that u is a homogeneous function of degree minus 1 because u of lambda x comma lambda y equals lambda power minus 1 times u of x comma y.
- After proving u to be homogeneous function, we can directly use Euler's formula, which is nothing but x into dou by dou x plus y into dou by dou y equals n u.
- Because n is minus 1, we can get this particular answer that is x into dou by dou x plus y into dou by dou y equals minus u becomes minus half x minus y divided by x square plus y square.
- Because n is minus 1, finally we get 2 times.
- X minus y divided by x square plus y square.
- U of x comma y becomes log u, then this part becomes x power 3 times y power 3 divided by x square plus y square.
- Then check whether f of u is homogeneous or not.
- After we simplify, we get lambda power x power 3 y power 3 divided by lambda square of x square plus y square.
- equals n times f of u by f dash of u.
- If u equal to log of x square plus y square divided by x plus y, then find a value of these two.
- Again observe carefully.
- Again, we use a different approach because log is involved, we take exponential on both the sides.

Example or application from the lecture

- After understanding the concept of homogeneous functions, let us solve problems on Euler's theorem.
- It is given that u equals e power x power 3 times y power 3 divided by x square plus y square, then find the value of x into dou u by dou x plus y into dou u by dou y using euler's theorem.
- Observe carefully, the given function is not homogeneous.

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EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 10:00–14:59

Then that leads to f of u equals a power u equals x square plus y square divided by x plus y.

Key details

- Then check whether f of u is homogeneous or not.
- or degree n equals to 1.

- X square dou square u by dou x square plus 2xy dou square u by doux dou y.
- Plus y square dou square u by dou y square equals g of u of g dash of u minus one.
- If u equal to sine inverse of x plus y divided by square root of x plus square root of y, then find the value of these two.
- After using simplification, we get f of u equals lambda power half times sin u.
- Plus one by two.
- In a similar manner, when we use the extension of Euler's theorem, we get this particular step.
- The right hand side is going to be half times tanue.
- And this part is going to be the derivative of half tan u, that is nothing but half second square u, and then we have minus one.

Example or application from the lecture

- Let us look at this particular problem.
- Again observe carefully, the given function is not homogeneous.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

12 Taylor's and Maclaurin's expansion

Study focus: Taylor's and Maclaurin's expansion. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000158 · EV0000159

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today you will be learning an important concept called the Taylor's and McLaren series, which are very helpful in reducing any complicated problems in the form of a polynomial, and it has a lot of applications in engineering.

Key details

- But if I don't let you use a calculator, you think that she has gone crazy and then, you don't know what's tan of cube root of three, and then on that I'm asking you to find exponential power of that.
- What goes behind that?
- Anything that is complicated.
- It will not understand your differentiation of sine, cos, or, if you want to give a function that's like e to the power of tan of x square and then ask you ask the calculator to differentiate it.
- What it does every time is it try to express any complex function as a series of polynomials.

- Few of them, I don't know, you might be aware because I think you are aware of binomial expansion.
- But if I make a small change in this, I'll give sine x square dx.
- Then what is the integral of this?
- Maybe if you want, I'll give you definite integral.
- Try plugging it this in the calculator because anyway you cannot find the value of this directly.
- Then the calculator writes this sine of x square in the form of a polynomial up to few terms, depending on the configuration of the calculator.
- And then it calculates once you get a polynomial, everything is easy, right?
- If I can write this as a polynomial, you integrate every term, apply the limits, simplify, you will get the answer.
- That is what is happening.
- Or as simple as I have mentioned here, e to the power of x square, which you don't know how to integrate.
- Zero by zero, and then you use L'opitholes rule, you did learn mean value theorems like Cauchy's and Legrange and Rolls.
- And you might be wondering how did they thought about those theorems?
- And m even though we are not discussing Taylor's theorem, but Taylor's series is nothing but consequence of Taylor's theorem.
- Why am I adding this continuous?
- Why am I prefixing this derivative with continuous?
- That's because I want to have a derivative of n plus one th order.
- I'm not just finding the first derivative, the second derivative, or the third derivative.
- In fact, I'm finding the n plus one order derivative.
- Then Taylor says that you can expand this function around any point A using a polynomial.
- So that means it looks like f of x, I can write this as f of a plus x minus a into f dash of a.

Example or application from the lecture

- suppose I'll ask you to,, find the value of this function, e to the power of tan of cube root of three.
- Another thing is suppose I'll ask you to evaluate this integral.
- Let's say I'll give you integral of sine of x.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 05:00–09:59

Plus x minus a the whole square divided by two factorial into f double dash of a and then this continues.

Key details

- This is not new to you, right?
- This is an infinite series, of course, but don't see that.

- Aren't you familiar with this?
- What is this?
- I can write rewrite this expression as f of x minus f of a divided by x minus a, which is nothing but f dash of a, right?
- Instead of expanding it around a point A, if I expand the function about the origin, about the origin, then we get the McLaren series.
- And once I have a polynomial, I can do everything with it.
- I can differentiate, I can study what is the change that is happening, I can integrate, I can find the limit of that.
- Whatever you ask me to do, I can do it.
- That's why these please remember that Taylor's in McLaren series always comes very handy for you when you have to handle complicated conditions or complicated functions.
- Mechlerin series, which you might already know, right?
- I to add a few, you can write.
- The series for this as well, which is 1 by 1 minus x.
- And one surprising thing is if they give me this series, I know that I'm getting a finite value.
- And that's very strange.
- Actually Ramanujan proved so many such results.
- Because you are adding infinitely many positive terms and it's coming back to minus one, which is not that easy to accept.
- what is happening there, right?
- But I want you to explore some of those things.
- I have log of 1 plus x, you can try log of 1 minus x, sine tan, hyperbolic, tan inverse.
- But how the question is, how are they getting this?
- It's very simple.
- I'll anyway, I won't be solving problems on Taylor's and McLaren series for one variable because I'll be extending it to bivariate function.
- But I can show you what happens.
- And I know that derivative of e par x is e par x itself.
- No matter how many times I differentiate, I'll get e power x only.
- And I can substitute these, replace this all these x's with 0.
- And that's not new to us, right?

Example or application from the lecture

- Suppose I'll show you why it is not new.
- Suppose I'll erase all the higher degree terms, okay?
- That means given any function, I can write this as a polynomial of infinite order with infinitely many terms.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

Core explanation · 10:00–14:59

All you need to do is find the value of the function at 0, provided f of 0 does not tend to infinity.

Key details

- I cannot write equal to because we cannot reach infinity.
- That's not defined because every at all the derivative at zero will tend to infinity.
- That's the only constraint.
- And here the same thing f of a cannot be infinite.
- Cannot tend to infinity or cannot be equal to infinite.
- And then I want to express it in terms of a polynomial.
- If you want about a point x is equal to a, then you use Taylor series, and if you want about the origin, you use metogen series.
- That's all you want, right?
- As I said before, it you can use this to evaluate any complicated function by expressing it in terms of a polynomial.
- You can evaluate definite integrals, which I also discussed, right?
- 0 to 1 sine of x squared.
- And if you want sine x square, just replace x with x square.
- That's all you need to do.
- Minus x to the power of 6 by 3 factorial plus x to the power of 10 by 5 factorial minus etc.
- And the question says you can you have to integrate this from 0 to 1.
- So that's easy now for me because I know what is integral of x square, x par 6, x par 10, etc.
- I can integrate all these things with respect to x and then apply the limits.
- That's all I need to do.
- I'll show you that when we plot sign any function and its series, okay?
- The next thing is understanding asymptottic behavior of the functions.
- that these are the asymptotes for the function, right?
- In this course, you will be learning higher order differential equations and when you need to solve the higher order differential equation, we will be basically using the Taylor series.
- And in physics, McLaren series used to approximate the Lorentz factor.
- And E is equal to M C square.
- The most famous equation E is equal to M C square.
- It's an approximation on velocities, which is done again by the McLaren series.
- In the motion of a pendulum, we approximate sine theta to be theta, that itself is a that means in the series expansion of sine theta.
- You're taking only one term.
- And again, it's used in power flow analysis of electrical power systems and can be used to find generating functions.
- This is a very, very important concept for computer science students.
- Okay, but anyway, when we do four-year series in the next semester, we'll talk about generating functions what it does and what do we mean by generating functions and other things.

Example or application from the lecture

- That means if of suppose I want to find the McLaren series for 1 by X.

- Suppose I have a rational function here and I want to check what is happening to its asymptotes.

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EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

Core explanation · 15:00–19:59

And the graph that is in red is the McLaren series for that.

Key details

- What is the series for sine x?
- Sine x is x by 1 factorial minus x cubed by 3 factorial and then plus x power 5 by 5 factorial and then etc.
- This blue graph is the graph of sine x.
- And then you see I have taken only four terms here.
- I have taken the maybe one more I'll add.
- in fact this is let me change this to blue so that I'm plotting sin x in blue.
- The next graph that you see.
- is again n is equal to 6.
- And see, this is maybe this is n is equal to 10.
- I'm just adding 10 terms in that polynomial, which is not too many as compared to an infinite series, right?
- It's overlapping.
- That's why it works so best.
- The first advantage is I'm getting everything in terms of a polynomial.
- That's the first advantage.
- I'm fine with that, I'm happy.
- The second thing is once it you are telling me to take infinite series, but most of the series converges for finitely many terms.
- Maybe not as m small as sinex, you might have to take say around 40 or 50 or whatever the number might be.
- That depends on the function.
- Again with the condition that this f of a comma b cannot go to infinity.
- plus this is f dash plus x minus a the whole square by two factorial into f double dash of n etc.
- I didn't talk much about this because you are already familiar with the binomial theorem, right?
- you already know what is binomial theorem expansion is and then how do you get the coefficients and other things.
- So for a Taylor series for a function of two independent variables will look like this f of ab plus x minus a into f x of a b because there are two derivatives.
- You have to take x minus a for fx, y minus b for f y.
- And there are four second order partial derivatives, fxx, fxy, and then f y.
- But that they are equal.

- F x y is the same as f yx.
- That's why you got this two.
- And then of course the next term will continue.
- The only thing I need to worry is not just this, but any of these cannot tend to infinity.
- I did mention only this but not necessarily.
- None of the derivatives fx, f y and f x, f y.
- At zero, zero, of course.
- At zero, zero.
- And it's easy to remember because fx goes with x.
- f y goes with y, f x goes with x square, fx y goes with 2xy I'm getting this too because mixed second order partial derivatives are equal and then f y goes with y square.

Example or application from the lecture

- But definitely the function matches exactly with the given function in terms of tailors or in terms of material series.
- Instead of AB I have to consider the origin.

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EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

Core explanation · 20:00–24:59

So,, before we solve some problems, this is the graph of polynomial because it's two dimension two variable I have to go to three d so it's not as good as visualizing in just one dimension.

Key details

- But anyway I have done my best.
- I have written a Taylor polynomial of this.
- so of just the first degree.
- and that's what I get.
- Maybe they have used the McLaren series.
- And now I'm taking the second degree polynomial.
- And then we'll see how many terms we need to take so that they're almost together.
- These are some of the applications of Taylor series for two variables.
- Basically, what they do is they just take a they are going to make use of the linear and optimization process.
- and distances can be measured accurately, variance of complicated configuration and as I said, visualization of complicated functions and so geometrically it's just that.
- In the next session, we'll try to solve some problems on Taylor's and McLaren series for a function of two independent variables.

Example or application from the lecture

- We can express any given function in terms of its polynomial.

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EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

13 Problems on Taylor's and Maclaurin's series

Study focus: Problems on Taylor's and Maclaurin's series. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000178 · EV0000179

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Alright, now let's try some problems on Taylors and McLaren series.

Key details

- As I was mentioning in the lecture, we are not going to solve problems on the Taylors and McLaren series for a function of single variable.
- But we'll try.
- And find out the Taylor series and the McLaren series for a function of two independent variables.
- So the first question is expand e power x cos y in powers of x minus 1 and y minus 4 up to second degree term.
- So the point is 1 comma pi by 4 because we want to express it in that form.
- x minus 1 into f x of 1 comma pi by four plus y minus pi by four into f y of 1 comma 5 by 4.
- So this is only the first degree term.
- The question says up to the second degree term.
- So the next one will be plus x minus 1 the whole square.
- And then plus etc.
- Of course, it's an infinite series.
- But what we are required to find is I need to calculate what is the value of the function.
- The function is e power x into cos y.
- And then we will find all the partial derivatives of this, or till whatever we require and the corresponding values.
- F of 1 is e and a cos of pi by 4 is root 2.
- Then what is f x of x y?
- F x of x y is you have to treat cos y as constant.
- What will be f y of x y now?
- I need to differentiate cos y by treating x as constant.
- Now we are done with the first order partial derivatives.
- F xx of x comma y will be I need to differentiate this again with respect to x.
- I need to differentiate fx with respect to y.
- And the last one that I need is f y f y of x comma y.
- So that means I need to differentiate f y with respect to y again.

- If you observe carefully, all these values are same and only thing they differ is because of the sign.

Example or application from the lecture

- So first let's write out what is the Taylor series expansion.
- And then now let's find the mixed second order partial derivative fxy of x y.
- That's because of the point that they have given about which we need to expand the function.

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EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 05:00–09:59

I can write f of x comma y as f of 1 comma pi by 4 is e pi root 2.

Key details

- So in fact I can take out e by root two common from all the terms because every term here has e by root 2.
- Next one is x minus 1.
- Plus, for y it is minus, so y minus y by four.
- This is the second degree term.
- Plus I have x minus one the whole square by two factorial.
- F x is positive.
- and minus x minus one of course into two into y minus pi by four divided by two factorial and the last term is also minus I'll get y minus pi by four the whole square divided by two factorial.
- So I can stop here for now because they have asked me to expand only up till the second degree term.
- So maybe in the next problem we'll see an expansion for Taylor series or McLaren series wherein we go up to third degree term.
- F of x y is equal to log of two x plus y plus one about the point zero comma zero up to three terms.
- That's a question.
- F x at zero coma zero by two factorial plus two x y.
- f x y 0 comma zero by two factorial plus f y at zero comma zero into y square by two factorial and then of course plus etcetera other terms will continue.
- I'll get log of 1, which is 0 for me.
- So first term, the constant term is not there for this particular function.
- f x will be 1 by 2x plus y plus 1 into 2.
- Because I am differentiating with respect to x, I need to treat the other two variables as constant.
- What will be FY of X, comma Y.
- If y is again 1 by 2x plus y plus 1 and the derivative of y is just 1.
- So f y of x comma y or 0 will be just 1 because these two terms goes to 0.

- So that means f x at 0 comma 0 will be minus 4.
- But let me find the two more terms, which is fx y.
- of x y so that I can complete all the second degree term.
- And the last one would be to calculate f y of x y.

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EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

Core explanation · 10:00–14:59

Will be the first term is 0.

Key details

- Okay, that's the first term.
- Minus 4x square by 2 factorial.
- Minus 2xy by 2 factorial.
- Actually these two terms were not necessary because the question says up to three terms.
- Two x plus y plus one will be two x minus or plus y minus two x square, okay, minus x y minus y square by two plus x.
- So the next problem we'll see it as an application of Taylor series.
- Now in the next problem we'll see how to apply this Taylor series to evaluate a value.
- This is the question, right?
- So that means just observe these values.
- Because I want the approximate value, can I say that f of 0.9 and minus 1.2 is approximately equal to f of 1 comma minus 1.
- And then so f x at 1 comma minus 1 will be you substitute y as minus 1 and this is square which is 2.
- Next is we'll find out what is f y of x comma y.
- But for now I think because we have three non zero terms, this should be sufficient.
- That means we got tan inverse of x y.
- I can write tan inverse of x y as minus pi by four plus I'm expanding about the point x minus one so x minus one into minus half plus y plus one into plus half.
- Plus, etcetera of course.
- But this also can give a better close approximation of this.

Example or application from the lecture

- Let's try another problem on find Taylor series expansion of f of x y equal to tan inverse of x y and find the approximate value of f of zero point nine comma minus one point two.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 15:00–19:59

0.9 and minus 1.2 will be minus 5 by 4.

Key details

- And if you substitute here for x is 0.9, minus 1 will be just 0.1 with a minus into minus half.
- Plus y is minus one point two plus one will be again minus zero point one minus zero point two in fact.
- Minus 0.1 by 2 is just the 1 by 20.
- This is just an application problem of how we can use the Taylor series to evaluate a function at a particular point.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

14 Extremas of functions

Study focus: Extremas of functions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000194 · EV0000195

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we'll be doing applications of partial differentiation, which is extremes of functions.

Key details

- I hope you remember how you calculated maxima for a function of single variable, right?
- And then we wanted to know what are these points, right?
- That's what we did.
- It could be either maximum or minimum.
- The first thing I did was to calculate the critical points.
- You calculate the first derivative and equate them to zero.
- What is the practical advantage of this?
- that if you have a local maxima, then the tangent at these points will be parallel to the x axis.
- Can you visualize that here?
- See k you can see that here.
- So we get the critical point by solving this equation that's the first step.
- The second step would be you've got to find the second order partial derivative.
- Second order not partial derivative, second order derivative.
- And if the value of this is less than zero,.
- At the critical point, we say that the function is maximum, right?
- And if it is greater than zero we say that the function has a minimum at that point.
- And if t square y by dx square is equal to zero, then it's neither maximum nor minimum.

- And the condition holds for both because the tangents as you see, the tangents at these points are also parallel to the x axis.
- So basically what we did you calculate the first order derivative, equate it to zero to get the.
- Critical points means the points at which the curve might have a extrema, either it could be a maxima or a minima.
- Then we need to decide whether that critical point represents a maximum point or a minimum point.
- If it gives zero then we cannot conclude anything.
- Such points are called saddle point.
- I hope you remember that, right so a saddle point looks something like this where so if saddle point is this where the function is neither maximum nor minimum.
- That means the test fails basically.
- But before that when do we say that a function of two independent variables has a maximum?
- I can decide that this point is the maximum for this surface at a point A, comma b if.
- Any neighborhood point the value of the function is less than the value of the function at A B.
- That means this condition.
- If f of ab is greater than f of a plus h, comma b plus k.
- Neighborhood means very close to this.
- Then we say that the function has a relative maximum.
- Why the word relative?
- It's because I'm considering the points that are closer to A B, I'm in a finite range, right?
- I'm not taking the surface in the entire real plane from minus infinity to infinity.
- That's why the term relative maximum.
- Because in that neighborhood that's the maximum point.

Example or application from the lecture

- Suppose that's the case.
- Or suppose I have a curve or a surface something like this.
- And let's say I have a point here and I call that as f of x comma y.

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EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then we say that's a relative minimum.

Key details

- So that means.
- In other words, you can have another definition if delta of A plus H comma B plus K minus F of A B is of the same sign, that's what I mean.
- So you just look at this condition here.
- For all small h and k and if sin is negative then f of ab is maximum.

- If it's positive we say that f of ab is minimum.
- That's the definition, okay.
- So you don't need to worry too much about it because what we call it as a maximum point and what we define it as a minimum point.
- So what we did first is we got dy by dx equal to 0.
- So, corresponding to this, what we have for a function of two independent variables, that there are two first order partial derivatives.
- So the first one to find the critical point I'm going to equate the first order partial derivatives to 0.
- And the second condition, what is the second condition here?
- I need to find.
- The second order derivative, substitute the point that you got here and check if the value is negative or if it's positive or if it's equal to C.
- That is, I am going to find because that there are four mixed order partial derivatives, right?
- R T minus a s square.
- Where what is r?
- R is nothing but f x.
- And so F y is also y x is also S.
- that mixed second order partial derivatives are always equal and FYY will be T.
- And this determinant is called a Hesian matrix.
- Is that's a condition I want.
- So remember here we got it directly, but here we need to check another condition.
- So the first preliminary condition that should be satisfied for a point to represent an extrema, it could be a maxima or a minima is f x into f y minus f x y the whole square must be greater than zero.
- If this R T minus S square is less than 0, then f of a b is not an extrema value at all.
- You cannot conclude anything from that.
- Just in case if it becomes equal to zero, then the test completely fails.
- You don't know whether you don't know it the you don't get any information about the nature of that point, whether it is a maxima or a minima or a saddle point, nothing we get.
- The first thing is find out both the first order partial derivatives, equate them to zero.
- Secondly, find R T minus S square, that is a combination of the second order partial derivative.
- At some point if this is zero, how do I decide that point is a maximum or a minimum.
- In that case, we have few conditions.
- Along with this condition, if R is less than 0 at any point AB, then f of A B has a maximum value.
- R means what then?
- This must be greater than 0.
- Along with this you need to check that f x must be less than 0.

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EV0000198 · EV0000199 · EV0000200 · +1 source units in pack.json

Core explanation · 10:00–14:59

which is R.

Key details

- So if this condition is satisfied then it will have a minimum or maximum.
- This is similar to the condition, see you remember this d square y by d x square has to be less than 0.
- And the second condition is so if R T minus S square this has to hold good for it to be an extrema.
- Along with that I have F Y now to be.
- sorry F x only.
- So, for a condition for it to be minimum now, right?
- Condition for it to be minimum is f x must be greater than 0 and that's what you concluded, right?
- In the case of single variable.
- If d square y by dx square is greater than 0, then the function will have a minimum at that point.
- And then lastly if I have R T minus S square to be less than 0, then it's it represents a saddle point.
- It's it doesn't sway to the left, neither it sway to the right.
- And of course if this is zero then I cannot conclude anything, so you have to say that no conclusion can be made.
- The first thing is you find out f x equated to zero, find f y equated to zero.
- Solve these two equations to get critical points.
- Next is you calculate the second order partial derivative.
- This has to be positive.
- Along with this, if f x is less than 0, I'll get maximum.
- That means that point represents maximum.
- And if x f x is greater than 0,.
- And if f x into f y minus fxy the whole square, if it is less than 0, then it's a saddle point.
- And if just in case if this is equal to zero then no conclusion.
- Is it clear?

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EV0000202 · EV0000203 · EV0000204 · +1 source units in pack.json

15 Extremas of functions continued

Study focus: Extremas of functions continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000206 · EV0000207

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Find the highest temperature on the surface of the unit sphere.

Key details

- because it's already in the xy plane.
- But do we know how to find the extremas for a function of three independent variables?
- That's a problem I have.
- I can write z square as 1 minus y square minus x square.
- x y into for z square I can substitute 1 minus x square minus y square.
- You differentiate that with respect to x partially and equate it to 0.
- I will get 400 y is a constant, I do not need to worry, derivative of x is 1, derivative of x cube is 3 x square and derivative of x y square is y square.
- If I equate this to 0,.
- Then I get 1 minus 3x square minus y square to be 0.
- That's a derivative.
- And if you equate this to zero you will get this equation.
- They are like simultaneous equations.
- you subtract them together.
- If you subtract I'll get minus two x square and then plus two y square equal to zero.
- If x is plus or minus y,.
- What w what is the value of x and y that I get if I substitute as x is equal to plus or minus y there.
- Let me substitute in either this equation or this equation, it doesn't matter.
- Or I get 1 is equal to 4 x square or x square equal to 1 by 4.
- But also what we got is y equal to 0 or this.
- Actually this should be x equal to 0, not y equal to 0.
- x is equal to minus half and y can be minus half.
- What we need to do is we just need to c get a combination of these things.
- You get these many points.
- Either it could be 0 half comma half, half comma minus half, minus half comma half and minus half comma minus half.
- That means these are the points at which there is a possibility for the function to have an extrema.
- I don't know whether it will be a maximum or a minimum or a saddle point, I don't know.
- So next thing what we need to do is we need to find the second order partial derivatives.
- And for half comma half it is positive, for minus half comma half also it is positive.
- And so you can calculate what is the maximum value that it can attain and minimum value it can attain by substituting these values, right.

Example or application from the lecture

- First thing is so that means they have given a sphere which might look something like that.
- And they have given the temperature function t of x y z as 400 x y z square which varies with the par with the spatial coordinates.

- So what I will do is so this is a constraint that is given to me, that means you need to find the temperature in space, find the highest temperature on the surface of the sphere.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000206 · EV0000207 · EV0000208 · +1 source units in pack.json

Core explanation · 05:00–09:59

maximum at two points.

Key details

- Can we conclude just by taking this?
- Just by taking this condition?
- I need to check along with that R is less than zero.
- I can erase the things that are there here.
- The function is this.
- We got F x already as four hundred y into one minus y square minus three x square.
- What is f x?
- I need to differentiate this which will give me minus 6x.
- Because it satisfies this condition, we say that the T is maximum at these two points on the surface of a sphere.
- And if you want to calculate what is that maximum value, substituting four hundred so we have four hundred into x.
- into y into z square.
- But z square is actually 1 minus x square minus y square.
- That means the maximum temperature that is attained on the surface of a sphere is fifty.
- So this is one method wherein we actually eliminate and then we substitute.
- This is more of a practical problem, but I felt like you will appreciate this because you will understand how to solve a problem.
- It says a container with an open top, let me underline whatever you need to be careful about is to have a ten meter cube capacity.
- and be made of thin sheet of metal.
- Calculate the dimensions of the box if it is to use the minimum possible amount of metal.
- So it's an open top means what?
- Because I am worried about the surface area.
- I hope it's clear why didn't I prefix this with two is because my top is open here as you can see this is not closed so because it's an open top box.
- That's a wall.
- I hope you understood how I wrote this.
- This is because I need to minimize the possible amount of metal that is required for me to made it is made out of thin.
- I cannot handle three independent variables, right?
- Like we did in the previous case, we use the equation of the sphere.
- We are going to eliminate z.
- I am doing little simplification here.

- So this is the function which I need to minimize because I want the minimum possible amount of metal to be used.
- Now what are the condition for this to be minimum?
- area is equal to 2xy plus 20 divided by x plus twenty divided by y and that's what we got.
- What are the conditions for us to minimize this?
- So the first thing is I need at what points it can be minimum.
- Ax will be 2 y minus 20 by x square, right?
- Equate this to 0.
- What do I get if I equate this to zero?

Example or application from the lecture

- So at any given point because of this negative, okay, if both x and y are positive, or both x and y are negative, I'll get f of x to be negative and that's what I want, right?
- Let's do another problem.
- And of course it's a rectangular box, so they have given XYZ to be ten.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000210 · EV0000211 · EV0000212 · +1 source units in pack.json

Core explanation · 10:00–14:59

I can take out y comma two comma and I can divide two out through out by two.

Key details

- Is that fine?
- So this is the first equity that I get y is equal to twenty by or ten by x square.
- What's a y?
- I will get 2x again minus 20 by y square equal to 0.
- If I equate this to 0, I will get x is equal to 10 by y square.
- Where should I put that to solve, right?
- I need to solve for x and then y and then z to get the critical points.
- The only equation that I know is XYZ equal to ten.
- I got x y is equal to ten by x square and x is equal to ten by y square.
- What is the next thing I know?
- I know that I have to find a x a y minus a x y the whole square must be greater than zero.
- Alright, so I have these two equations now y is equal to 10 by x square and x is equal to 10 by y square.
- Can I substitute one in another?
- I have y is equal to 10 by x square.
- Or I can eliminate this.
- Can we solve this?
- I'll get y minus y to the power of four by ten to be equal to zero.
- Is it fine?
- So that means I can take out y common and then I get 1 minus y cube by 10 equal to 0.
- What do we get when we solve this equation?

- I'll get y equal to 0.
- as the first value or so the other equation that I have is 1 minus 1 minus y cube by 10 equal to 0.
- Y is equal to zero doesn't make sense for me.
- And so how do I find x now?
- I know that x is equal to ten by y squared.
- And of course the y equal to 0 doesn't work because if y becomes 0 x tends to infinity.
- That's another way to eliminate that.
- And how do I get z?
- Z is equal to 10 by x y.
- This is 10 divided by 10 to the power of 2 by 3 or this is 10 to the power of 1 minus 2 by 3 which is 10 to the power of 1 by 3 again.
- Is it fine?
- And the we just substituted that.
- We got x is equal to y is equal to z.
- But how do we check that this will m this is the minimum value or the minimum amount of metal that I can get, right?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000214 · EV0000215 · EV0000216 · +1 source units in pack.json

Core explanation · 15:00–19:59

20 by x square and derivative of 1 by x square will be 2 by x cube with a minus.

Key details

- And a y will be same thing, which is 40 by y cube.
- And then I need another axy.
- What are the condition we have?
- Ax x into ay minus a x y the whole square.
- And what happens to this?
- A x we got the value right.
- What is the value that we got?
- 40 by x cube.
- So that means this is greater than 0.
- And what is Ax?
- You can observe that for all the points that we got for x, y and z, a x is always greater than 0.
- That means the dimensions of the box so that the minimum of metal sheet is used will be x is equal to cube root of ten.
- It's a basically a cube and y is equal to cube root of ten and z is equal to cube root of ten.
- I hope the problem is clear.
- in the next session we will see another method wherein we don't have to eliminate.
- See the problem with us we had like we have to eliminate one of the variables to make it into a function of two independent variables.

- It might not be as simple as this in every case, right.
- So we'll see another method which will help you to minimize that so that how to solve the problem.

Example or application from the lecture

- So, the condition that we got we got the critical points now let's check the condition.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000218 · EV0000219 · EV0000220 · +1 source units in pack.json

16 Problems on Maxima and Minima for a function of two variables

Study focus: Problems on Maxima and Minima for a function of two variables. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000222 · EV0000223

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we solve problems on maxima and minima for a function of two variables.

Key details

- F of x comma y.
- Find the maximum and minimum values of function x power 3 plus 3xy square minus 15x square minus 15y square.
- As per the procedure, we first find out the partial derivative of f with respect to x and y, which we call them as p as well as q.
- Next, in order to find out critical points, we need to equate fx as well as f y to zero.
- From f y equal to zero, we see that either y equal to zero or x equal to five.
- It means when we equate 6x minus 30y to 0, we see that either y 0 or x equal to 5.
- We have one more case left out.
- In a similar manner, we have to put x equal to 5 in expression number 1.
- Then we need to compute R S t.
- R is nothing but dou square f divided by dou x square, which is 6x minus 30.
- Then S is nothing but dou square f divided by dou x dou y which is nothing but 6y.
- So therefore R T minus s square becomes 36 of x minus 5 whole square minus pi square.
- As we see that there are four critical points.
- So we can think of four cases because of four critical points.
- At 0, 6 comma 0 we see that R t minus s square is positive, but R is positive.
- Therefore, we can say that at 6 comma 0 the function is minimum.

- At the critical point 4 comma zero, what happens?
- At 4 comma zero, R T minus S square is 36, which is positive, but R is negative.
- Therefore, we say that at the point 4 comma 0, the function is maximum.
- Therefore we say that we say that these are the two stattle points.

Example or application from the lecture

- Let's start with this problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000222 · EV0000223 · EV0000224 · +1 source units in pack.json

Core explanation · 05:00–09:59

We first found out PQ, then we equated P and Q to 0.

Key details

- Then we found out critical points.
- Then we computed RT minus S square.
- Then evaluated RT when C square at each of those critical points then arrived at this conclusion.
- Find the highest temperature on the surface of the sphere x square plus y square plus z square equal to a square.
- Note that because we are looking for maximum temperature, we assume that k to be a positive constant.
- We need to convert this function of three variables to a function of two variables by using z value.
- P of x y z becomes k x y of a square minus x square minus y square.
- The value of R is going to be minus 6K XY.
- The value of S is going to be K of A square minus 3x square minus 3y square.
- The value of T is going to be minus 6 Kxy.
- Therefore, we say that 0.0 is a saddle point.
- In a similar manner, we see that at the points a by 2, a by 2, as well as minus a by 2, and minus a by 2, R T.
- Minus S square is positive, but R is negative because we are assuming K to be positive.
- Test the function for maxima, minima and settle point.
- Fy is going to be 4y power 3 minus 2y.
- Therefore, we say that the function is maximum at the point zero comma zero and the maximum value is one.
- In a similar manner, it is easy to see that at all these four points, R T minus S square is positive, but R is also positive, therefore we say that F is minimum at these points.
- Lastly, at plus r minus 1 by root 2 0, and 0, plus r minus 1 by root 2, we can see that r t minus s square is minus 8, which is negative, but r is also negative.
- Therefore, we say that these are the two saddle points.
- F of x comma y to be sine x plus sine of y plus sine of x plus y.

Example or application from the lecture

- Let's look at another problem.
- The temperature capital T at any point XYZ in space is given by this expression KXY Z square where K is a constant.
- Let's evaluate R T minus S square at each of these critical points.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000226 · EV0000227 · EV0000228 · +1 source units in pack.json

Core explanation · 10:00–14:59

Fy is going to be cos y plus cos of x plus y.

Key details

- Subtracting 2 from 2, we see that x equals y after cancellation of plus of x plus y and plus of x plus y.
- Then substitute x as y in expression 1, we see that x equal to 5 by 3.
- So because x is y.
- the only critical point is pi by three comma pi by three so at pi by three comma pi by three we see that after evaluat r t minus s square is positive but r is negative therefore we say that.

Example or application from the lecture

- The function is maximum at the point pi by three, comma pi by three, the maximum value is given by three root three divided by two.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000230 · EV0000231 · EV0000232 · +1 source units in pack.json

17 Lagrange's method of undetermined coefficient

Study focus: Lagrange's method of undetermined coefficient. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000234 · EV0000235

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In the previous session we discussed about the extrema of a function of two independent variables, right?

Key details

- And we did solve some problems also.

- Now we are going to study another method which is called the Lagrangius method of undetermined multipliers.
- So such problems wherein we get a val the function along with the constraint can be solved by using Lagrange's method of undetermined multiplier.
- The next thing that is you need to do is you solve the first order partial derivatives.
- This is an advantage.
- You remember what we did here?
- We had x square plus y square plus z square equal to 1.
- So this z square we substituted back here and then eliminated that z that can be avoided if we use this method.
- That's why it's called an multiplier because this is like a scaling of the constraint, right?
- And it is undetermined, so we don't need to worry about what is the value of that lambda, but we'll calculate the value, okay?
- So calculate the first order partial derivatives, equate it to zero and use the constraint.
- Constraint is phi of x y z.
- for to solve lambda x y and z.
- Three of them I got here f x 0, f y 0 and f z 0.
- I'll I'll use that constraint and then I'll s use that to solve the lambda value.
- The stationary points are nothing but the critical points, right?
- You remember what we did in the previous case?
- Ah and second advantage is this method can be extended to n parameters, right?
- If I have x of x y z w, then you write the equation by introducing lambda, that's all you have to do.
- But the disadvantage is you will not be able to know what is the nature of the stationary point or what is the nature of the critical point.
- That means you will not be able to decide whether that will represent a maxima point or a minimum point, that is the disadvantage.
- All you can do is get the stationary points.
- that it will represent either maximum or minimum.
- But I cannot verify if I have to verify this R T minus S square to be greater than 0 and R is less than 0 or R is greater than 0, I have to go back to the previous method only.
- That's the disadvantage of this.
- But definitely to find the critical point you can make use of this.
- So that means that might look something like let me try to sketch that.
- I don't know how well I can sketch.

Example or application from the lecture

- So, in the first problem if you recollect what we did is we studied the temperature if the temperature function was XYZ was 400 XYZ square, this was given to us, and then we had to calculate what is the maximum temperature on the surface of a sphere.
- from the given function.
- Let's try a problem, maybe things will be little more clear, right?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000234 · EV0000235 · EV0000236 · +1 source units in pack.json

Core explanation · 05:00–09:59

This so I have the sphere.

Key details

- And then now what I want is I have to inscribe a rectangle in it.
- If this has to have a maximum value, then it has to be a cube.
- It's x square plus y square plus z square equal to r square with the center of course at the origin and radius r.
- And what is the volume of the rectangular solid, F of X Y Z?
- Two eight x.
- That's because I have taken two x and then two y and two z.
- Which one I need to maximize or minimize, I need to worry about this.
- What is the constraint inside the sphere.
- The volume is like length into breadth into height, so you can take that as x y z.
- And what is FY?
- F y will be x z plus lambda into f y so I have to I'll get two y.
- Plus lambda into two set.
- I can take out z common and then I get lambda y plus two lambda plus y equal to zero.
- Don't take this 8 also it is absolutely fine.
- But what you can do is if you multiply this first equation with x, I will get x y z plus 2 x square lambda is 0.
- I will do the same thing for the next equation.
- I will take x y z plus to y square lambda equal to 0.
- What do I get if I solve these two equations?
- I will subtract them.
- I will get 2x square lambda minus 2y square lambda equal to 0.
- This will give me lambda equal to 0 or x is equal to plus or minus 1, right.
- Because it's a measure minus y does not matter for me, so I will get x is equal to y.
- We never use the fact that I have to take R T minus S square and then find out R greater than zero or less than zero.
- In such problems the Slagrange's method of undetermined multipliers are very helpful.
- The question is find the points closest to the origin of the hyperbolic cylinder.
- X square plus Y square plus Z square because a sphere can be taken as a point.
- Maybe if I take this as R instead of that.
- As R becomes R square maybe.

Example or application from the lecture

- Let's prove that mathematically, okay.
- That was not necessary because the question said that maximum volume, they have already given that it is maximum.
- Let's try another problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000238 · EV0000239 · EV0000240 · +1 source units in pack.json

Core explanation · 10:00–14:59

If R becomes zero, I will get the s smallest a point which is nothing but the origin.

Key details

- On the hyperbolic cylinder, x square minus z square equal to.
- Or if you want to call g of x y z, g of x y z is x square minus z square minus 1.
- I need to find XY Z and that lambda.
- First let me calculate what's f x is.
- F x will be two x plus two x lambda, right?
- and f y will be again two y plus a zero, there is no y term there, and f z will be two z minus two lambda z.
- But I have another condition right which is x square minus z square minus one equal to zero.
- I have to equate them to zero, right?
- That's that's what is the condition, f x is zero, f y is zero and f z is zero.
- What about the first equation if you observe carefully the first equation is 2x plus 2x lambda is 0.
- This gives me x equal to 0 or lambda is equal to minus 1.
- But can x be zero?
- There is a problem with that.
- If x becomes zero, if x is zero, then what do I get?
- I'll get z square to be minus one.
- So that means my z will not be in the real plane at all.
- That's a problem I have, right?
- So that means x is equal to zero is not possible.
- But I can take definitely lambda equal to minus one.
- Now what is the next equation I have?
- I have two z.
- Ah 2 z minus 2 lambda z equal to 0.
- This will give me z equal to 0 or lambda equal to 1.
- That is fine for me.
- But if z is 0 what is the value of x that I get?
- I have x square minus z square minus 1 is 0.
- That means the point that will be very close to the origin will be plus or minus 1, 0 and 0.
- Is it clear?
- So you remember if you are using Lagrangius method of multiplier, the condition will also be mentioned.
- That means that point is minimum only for you.
- Here it is closest.
- What is the rectangular solid of maximum volume that can be inscribed in a sphere.
- So whenever you have a constraint then you can definitely make use of Lagrange's method of undetermined multiplier, that's one thing.

- Similarly the first problem that we did in the previous session can also be solved using Lagrangian's method of undetermined multipliers.
- You practice some more problems then concept should be more clear to you, right?

Example or application from the lecture

- My axillary equation would look like f of x y z is equal to this is the given function x I want to make this as small as possible with the given condition that x square minus z square minus.
- In the previous example it was maximum right.
- The second thing is whenever they have already given the condition that you have to maximize the temperature.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000242 · EV0000243 · EV0000244 · +1 source units in pack.json

18 Problems on Lagrange's method of undetermined multipliers

Study focus: Problems on Lagrange's method of undetermined multipliers. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000246 · EV0000247

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will small problems on Lagrange Method.

Key details

- Find the dimensions of the rectangle box open at the top of maximum capacity whose surface is 432 square centimeter.
- Two XY plus two Y Z plus two Z because the boxes open at the top.
- Because we need to find out the dimensions that is XYZ of this rectangular box.
- This volume xyz becomes function for the problem and xy plus 2yz plus 2 z equals 432 becomes constraint for the problem.
- plus 2 zx minus 432 equals 0.
- That is xy plus two yz plus two zx minus four thirty two.
- Next, as per the procedure, we need to compute the partial derivative of f with respect to x, y, as well as z.
- Then we equate fx to zero.
- That leads to yz lambda of y plus two z equals zero.
- of f with respect to y equals zero gives x plus lambda x plus two z equals zero.

- We see that 2z lambda of x minus y equals 0.
- Because 2z lambda is not zero, we can equate x minus y equals 0, then that gives x equals y.
- Similarly, multiply 2 by y and 3 by z and subtracting we get x lambda of y minus 2 z t equals 0 because x lambda not zero we can equate y minus 2 z to 0 that gives y equal to 2 z.
- Substitute x equals y, y equals to z in the constant and for you have x y z.
- We see that.
- 3x square equals 432 or x equals 12.
- Because y is x, so y is 12.
- Because z is y by 2, z becomes 6.
- So this is how we can solve problem using Lagrange method.

Example or application from the lecture

- So let's look at the first problem.
- Let us use the grandiose function which is given by f of x y z equals f of x y z plus lambda times phi.
- Let us use this in the constraint.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000246 · EV0000247 · EV0000248 · +5 source units in pack.json

Core explanation · 05:00–09:59

Let's look at the second problem.

Key details

- Divide the number 24 into three parts such that the continued product of the first square of the second and the cube of the third may be maximum.
- Continued product means the product we get by multiplying together in succession.
- Then divide the number 24 into three parts.
- It means we want to split 24 into three non-negative parts.
- X plus Y plus Z minus 24.
- F to zero gives two x y z power three plus lambda zero.
- Observe carefully in all these three expressions, we can use this idea.
- In each case, lambda equals minus y square z power 3, minus 2x y z power 3, minus 3xy square z square.
- We see that minus y square z power 3 and minus 2x y z power 3.
- It means from the first two terms we can get y equals to 2x.
- So because y is 2x, y becomes 8, and because z is 3x, z is 12.
- It means the three parts are 4 and 8 and 12.
- Because this is the only condition under which the inscribed rectangular solid achieves maximum volume inside the sphere.
- And f z to 0, so we see that lambda equal to minus y z by 2x, lambda equal to minus x z by two y, lambda equals minus x by 2z.
- When we solve all these in ratios, we see that x equals y equals a.

Example or application from the lecture

- Before we solve this problem, let's understand what is the meaning of continued product.
- It means if we have three parts, let us say x comma y comma z, then the continued product of the first square of the second and the cube of the third is going to be x times y square times z power three.
- Let's say xyz, that is x plus y plus z equals 24.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000250 · EV0000251 · EV0000252 · +1 source units in pack.json

19 Source slide deck: Introduction to Maxima and Minima for a function of two variables

Study focus: Source slide deck: Introduction to Maxima and Minima for a function of two variables. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000660 · EV0000661

SHORT-NOTE RECAP

Study notes · Source slide deck: Introduction to Maxima and Minima for a function of two variables

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · [Image text, OCR]

- Department of Science and Humanities

Slide 2 · ENGINEERING MATHEMATICS - I

- Sub Topic : Maxima and Minima of a Function of Two Variables

Slide 3 · ENGINEERING MATHEMATICS - I

- Maxima and Minima for a Function of Two Variables
- These are all local maxima

Slide 4 · ENGINEERING MATHEMATICS - I

- Maxima and Minima for a Function of Two Variables
- These are all local minima

Slide 5 · ENGINEERING MATHEMATICS - I

- Maxima and Minima for a Function of Two Variables

Slide 6 · ENGINEERING MATHEMATICS - I

- A function $f(x, y)$ is said to be have a relative maximum at a point (a, b) if there exists a neighborhood of the point (a, b) say $(a + h, b + k)$, $h\ and\ k$ are small such that $f(a, b) > f(a + h, b + k)$.
- If $f(a, b) < f(a + h, b + k)$ then $f(x, y)$ is said to have relative minimum at (a, b). Also $f(a, b)$ is said to be an extreme value of $f(a, b)$ if it is either a maximum or a minimum.
- In other words if $\Delta = f\ a + h, b + k - f\ a, b$ is of the same sign for all small values of h, k and if this sign is negative then $f(a, b)$ is maximum & if this sign is positive then $f\ a, b$ is a minimum.

Slide 7 · ENGINEERING MATHEMATICS - I

- The necessary and sufficient condition for $f(x, y)$ to have a maximum value at (a, b) is $f\,x\ a, b = 0$, $f\,y\ a, b = 0$ and
- Note: If $rt - s2 < 0$, $f(a, b)$ is not an extreme value. i.e
- If $rt - s2 = 0$, then the test provides no information about the nature of the point.

Slide 8 · ENGINEERING MATHEMATICS - I

- Maxima and Minima for a Function of Two Variables
- If $rt - s2 > 0$ and $r < 0$ at
- If $rt - s2 > 0$ and $r > 0$ at value.

Slide 9 · ENGINEERING MATHEMATICS - I

- Maxima and Minima for a Function of Two Variables
- 5. If $rt - s2 < 0$ at
- 6. If $rt - s2 = 0$ at a, b the case is doubtful and needs

Slide 10 · [Image text, OCR]

- Department of Science and Humanities

EV0000660 · EV0000661 · EV0000662 · +27 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Continuity and differentiability of a function of two independent variable transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Continuity and differentiability of a function of two independent variable transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Continuity and differentiability of a function of two independent variable transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Continuity and differentiability of a function of two independent variable transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to Partial Differentiation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Introduction to Partial Differentiation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to Partial Differentiation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Introduction to Partial Differentiation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Geometrical interpretation of Partial derivatives transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Geometrical interpretation of Partial derivatives transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Geometrical interpretation of Partial derivatives transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9990; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Geometrical interpretation of Partial derivatives transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Total derivatives, Chain rule transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Total derivatives, Chain rule transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Total derivatives, Chain rule transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Problems on Total derivatives, Chain rule transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Total derivative, Chain rule and Implicit functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Total derivative, Chain rule and Implicit functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Total derivative, Chain rule and Implicit functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Total derivative, Chain rule and Implicit functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Implicit functions and problems transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Implicit functions and problems transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Implicit functions and problems transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Implicit functions and problems transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Composite and Implicit functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Composite and Implicit functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Composite and Implicit functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Problems on Composite and Implicit functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Homogeneous functions and Euler's theorem transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Homogeneous functions and Euler's theorem transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Homogeneous functions and Euler's theorem transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
Homogeneous functions and Euler's theorem transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's Theorem transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_Theorem__6c4797f9ed9a5b61.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's Theorem transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_Theorem__6c4797f9ed9a5b61.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Euler's Theorem transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_Theorem__6c4797f9ed9a5b61.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's Theorem transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_Theorem__6c4797f9ed9a5b61.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's theorem transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's theorem transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's theorem transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's theorem transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Taylor's and Maclaurin's expansion transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Taylor's and Maclaurin's expansion transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Taylor's and Maclaurin's expansion transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Taylor's and Maclaurin's expansion transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Taylor's and Maclaurin's series transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Taylor's and Maclaurin's series transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Taylor's and Maclaurin's series transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9983; only the JSON segmentation supplies the displayed timestamps.
Problems on Taylor's and Maclaurin's series transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Extremas of functions continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Maxima and Minima for a function of two variables transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Maxima and Minima for a function of two variables transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Maxima and Minima for a function of two variables transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9962; only the JSON segmentation supplies the displayed timestamps.
Problems on Maxima and Minima for a function of two variables transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagrange's method of undetermined coefficient transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagrange's method of undetermined coefficient transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagrange's method of undetermined coefficient transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Lagrange's method of undetermined coefficient transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Lagrange's method of undetermined multipliers transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Lagrange's method of undetermined multipliers transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Lagrange's method of undetermined multipliers transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9967; only the JSON segmentation supplies the displayed timestamps.
Problems on Lagrange's method of undetermined multipliers transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Slide page text: Introduction to Partial differentiation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Introduction to Partial differentiation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Introduction to Partial differentiation (converted PDF; original hash a1c216d29db4461b50640f770f898294307495d0f4ecabc6d0c546abd126a64d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Geometrical interpretation of Partial derivatives Problems on partial derivatives	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Geometrical interpretation of Partial derivatives Problems on partial derivatives	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Geometrical interpretation of Partial derivatives Problems on partial derivatives (converted PDF; original hash 4df6156d5226cd03f8889623b0ef2a5bde035e033d9b3ce84064ba9360944efc)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Total derivative Chain rule problems	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Total derivative Chain rule problems	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Total derivative Chain rule problems (converted PDF; original hash 8fb90d8c86fdd0e6672954f6874fa5df34e8e2988f0f7bf6085fd3b0b27ca5c6)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Implicit functions Problems on Implicit and Composite functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Implicit functions Problems on Implicit and Composite functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Implicit functions Problems on Implicit and Composite functions (converted PDF; original hash 33599a90d72d93b72a632725fa279ebbdd8da8def5fd3e5c8e8524224ba3373d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Homogeneous functions and Euler s theorem	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Homogeneous functions and Euler s theorem	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Homogeneous functions and Euler s theorem (converted PDF; original hash 5e9df5fbab859c24989026e80d8ec451101874e569137c4a4fb0789f4a16fd53)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Problems on Euler s theorem	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Problems on Euler s theorem	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Problems on Euler s theorem (converted PDF; original hash 84ce9a0df0c6f69428b12df247d69c4d2609eecefdb0ef07aca0cba7ab50bfa7)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Taylor and Maclaurin series expansion for a function of two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Taylor and Maclaurin series expansion for a function of two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Taylor and Maclaurin series expansion for a function of two variables (converted PDF; original hash 90064a7ee2a213d4017032a6ebaf80ebc3d8d90aaa7ffd21b26db378c85946ef)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Taylor s and Maclaurin s expansion for a function of two variables continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Taylor s and Maclaurin s expansion for a function of two variables continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Taylor s and Maclaurin s expansion for a function of two variables continued (converted PDF; original hash 9c39f5b326ed0806d84b6995f4f749e9afaf10afd48666cd3bcb52027f2ea4b2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Introduction to Maxima and Minima for a function of two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Introduction to Maxima and Minima for a function of two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Introduction to Maxima and Minima for a function of two variables (converted PDF; original hash e02458424ddafb4c04372fdf8ab6714366a7cf827e1ca9a819bbcc4c16e56cbe)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Problems on Maxima and Minima for a function of two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Problems on Maxima and Minima for a function of two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Problems on Maxima and Minima for a function of two variables (converted PDF; original hash 13b0c81bb2e7cdd3ac55cdaaa4466dad1e67d1b8b7d9b89827aac3734fffb57f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Lagrange s method of un determined multipliers for two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Lagrange s method of un determined multipliers for two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Lagrange s method of un determined multipliers for two variables (converted PDF; original hash 80306ba85a4b394f5259f1d9bcbc45bd8dd670f103fbdbc479dd0d1504523e5f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.
SRC00002	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00003	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00005	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00006	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00007	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00009	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00010	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00011	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00013	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00014	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0045. Sidecar variants compare...
SRC00015	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0045. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0045. Sidecar variants compare...
SRC00017	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0045. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00018	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0059. Sidecar variants compare...
SRC00019	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0059. Sidecar variants compare...
SRC00021	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00022	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00023	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00025	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00026	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0042. Sidecar variants compare...
SRC00027	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0042. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00028	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0042. Sidecar variants compare...
SRC00029	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0042. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00034	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0034. Sidecar variants compare...
SRC00035	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0034. Sidecar variants compare...
SRC00036	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0034. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00037	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0034. Sidecar variants compare...
SRC00038	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0048. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0048. Sidecar variants compare...
SRC00040	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0048. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0048. Sidecar variants compare...
SRC00042	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00043	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...
SRC00044	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00045	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...
SRC00046	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00047	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...
SRC00048	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00049	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...
SRC00050	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00051	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare...
SRC00052	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00053	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare...
SRC00054	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00055	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...
SRC00056	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00057	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...
SRC00058	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0039. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00059	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0039. Sidecar variants compare...
SRC00060	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0039. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00061	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0039. Sidecar variants compare...
SRC00062	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00063	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00064	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00065	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00066	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0036. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00067	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0036. Sidecar variants compare...
SRC00068	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0036. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00069	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0036. Sidecar variants compare...
SRC00070	0 covered · 0 excluded · 18 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00072	0 covered · 0 excluded · 18 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00071	0 covered · 0 excluded · 18 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00073	0 covered · 0 excluded · 11 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00075	0 covered · 0 excluded · 11 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00074	0 covered · 0 excluded · 11 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00076	0 covered · 0 excluded · 28 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00078	0 covered · 0 excluded · 28 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00077	0 covered · 0 excluded · 28 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00079	0 covered · 0 excluded · 18 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00081	0 covered · 0 excluded · 18 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00080	0 covered · 0 excluded · 18 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00082	0 covered · 0 excluded · 8 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00084	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00083	0 covered · 0 excluded · 8 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00085	0 covered · 0 excluded · 18 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00087	0 covered · 0 excluded · 18 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00086	0 covered · 0 excluded · 18 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00088	0 covered · 0 excluded · 13 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00090	0 covered · 0 excluded · 13 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00089	0 covered · 0 excluded · 13 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00091	0 covered · 0 excluded · 20 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00093	0 covered · 0 excluded · 20 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00092	0 covered · 0 excluded · 20 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00094	10 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00096	10 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00095	10 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00097	0 covered · 0 excluded · 15 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00099	0 covered · 0 excluded · 15 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00098	0 covered · 0 excluded · 15 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00100	0 covered · 0 excluded · 15 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00102	0 covered · 0 excluded · 15 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00101	0 covered · 0 excluded · 15 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

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